SIEMENS EDA

# HyperLynx™ Schematic Analysis Model Library Guide and Reference

Software Version 2604

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SIEMENS

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# Table of Contents

# 1. Creating HyperLynx Schematic Analysis Models

An intelligent model characterizes a device for use in schematic analysis with the HyperLynx Schematic Analysis tool.

You create intelligent models using the specifications listed from manufacturer datasheets. You enter device information from datasheets into model templates to standardize the data format while allowing customization to reflect specific programming and configuration.



## Prerequisites for Creating a Model

Before you create models and add them to your library, you need a project folder, bill of materials (BOM), schematic netlist, and datasheets for each component in the netlist.

For information about setting up the project (the project directory and its contents) and importing the bill of materials and schematic netlist, see "Setting Up a Project" in the *HyperLynx Schematic Analysis Guide and Reference*. When you import the bill of materials into the HyperLynx Schematic Analysis tool, the tool reports to you which components do not have models in the model library. You must create models for these components.

Before you can make a model of a component, you must have a datasheet of the component. To help you identify the correct datasheets, the manufacturer part number for each component is in the bill of materials. The following suggested websites have datasheets online:

- www.octopart.com
- www.datasheets360.com
- www.alldatasheet.com
- www.digikey.com

After you locate the required datasheets, review the ordering information section of each datasheet to confirm that the manufacturer and manufacturer part number matches the specifications in the bill of materials. Save the latest datasheets into the *Datasheets and Models* directory located in the project directory.

> **Tip**
>
> To interpret the functionality and electrical specifications of a datasheet, refer to *Understanding and Interpreting Standard-Logic Data Sheets* by S. Nolan and J. Soltero from Texas Instruments™.

# HyperLynx Schematic Analysis Intelligent Model Template

The HyperLynx Schematic Analysis tool reads component specifications from models that you create based on information you get from component datasheets.

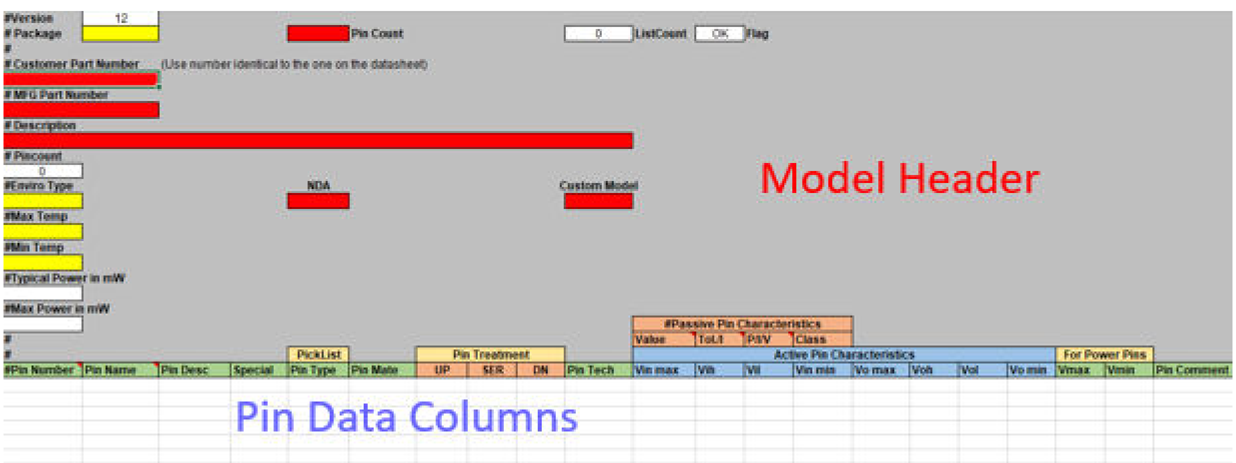
The intelligent model template uses conditional formatting, embedded macros, and dropdown lists to provide organization and an easy workflow process when modeling. You can use either Microsoft® Excel® 2013 or the equivalent Open Office tool.

The HLSA model templates are provided on Support Center at https://support.sw.siemens.com/en-US/product/1130580441/downloads/related/62D3qg7DUD4RW1T4WBInnv

## Intelligent Model Template

Use the intelligent model template to model active and passive components. An active component is a device that requires an external source to operate or generates power internally. Active devices range from discrete devices such as transistors to more complicated circuits such as digital and analog integrated circuits.

**Figure 1-1: Intelligent Model Template**




There are two main sections in the model template:

- **Model header** — Describes the generic device information.
- **Pin data columns** — Specifies the pin information of the device.

**Caution**

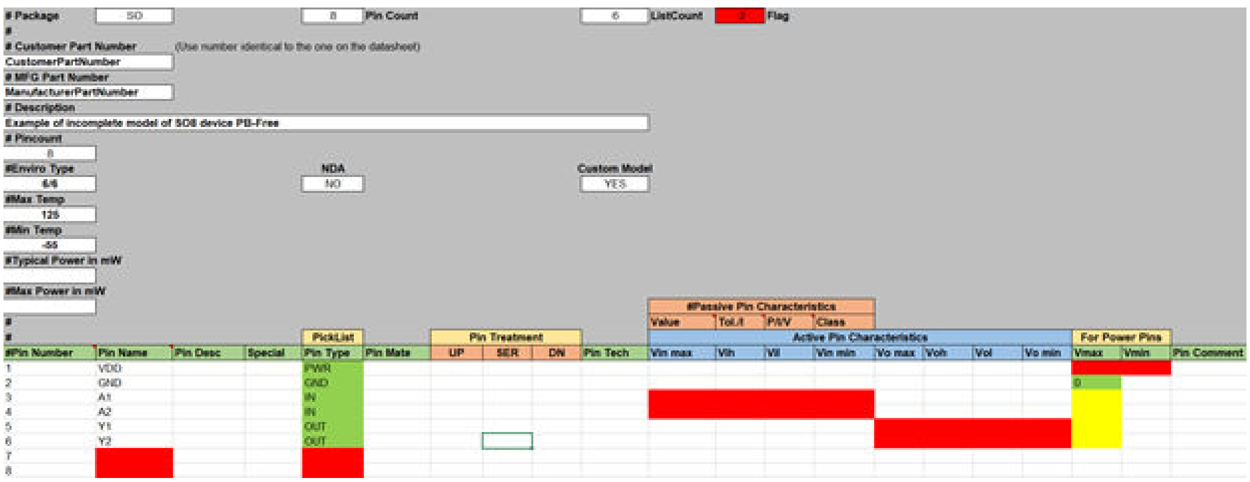
Do not alter the template in any way, except for adding information in the permitted cells. For example, do not remove any pound signs (#) or the model will not import correctly.

The figure shows an example of a simple four-pin oscillator. You create a model by entering the information required for each cell.

**Note**

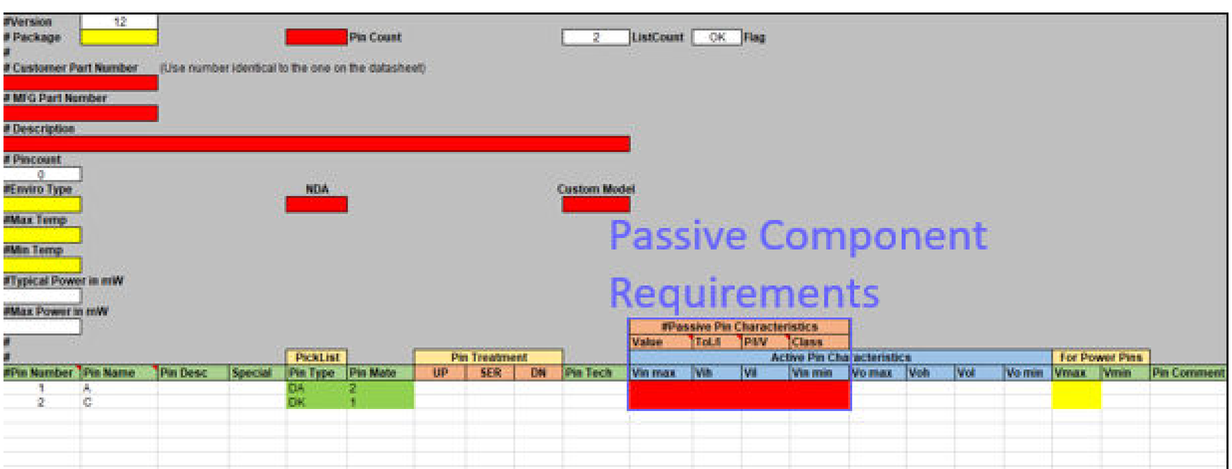
The Pin Type column includes a dropdown list of the available pin types.

**Figure 1-2: Example Active Model**




You can also use the model template to model discrete passive components such as diodes, resistors, inductors, and capacitors.

**Figure 1-3: Example Passive Model**



The model template automatically displays a half finished model of a two-pin diode. If the device you want to model is not a two-pin diode, you can replace this default information by selecting the information under the pin data columns and pressing the Backspace key.

## Conditional Formatting

The model template uses conditional formatting in the form of color coding to emphasize mandatory and optional information. The table summarizes the conditional formatting for the model header and pin data columns.

**Table 1-1: Conditional Formatting Rules for the Model Template**

| Model Header: | Before | After | Pin Data Columns | Before | After |
|---|---|---|---|---|---|
| Mandatory | Red | White | Mandatory | Red | Green |
| Optional | Yellow | White | Optional | Yellow | Green |
| Special | Red | White | Duplicate | White | Yellow |

The Before and After columns refer to the color of the cell before and after you enter data. In the model header, the NDA and Custom Model fields are considered as Special (see **Model Header**). In the pin data columns, if you enter the same Pin Number/Name in more than one row, all of the Duplicate cells instantly emphasize in yellow.

## Invalid Characters

The HyperLynx Schematic Analysis tool does not allow for most non-numeric or non-alphabetical characters with the exception of underscores (_). When you create the model, it is important to replace all of the invalid characters in the model with the replacements shown in the table.

**Table 1-2: Invalid Characters in the Model Template**

| Invalid Character | Replacement | Notes |
|---|---|---|
| Asterisk (*), Braces ({}), Parenthesis (()) | Remove All | Square braces ([]) are valid |
| Comma (,), slash (/), backslash (\) | Underscore (_) | |
| Plus sign (+) | Lowercase or uppercase P | Usually the plus sign (+) in a differential pair |
| Minus sign (-) | Lowercase or uppercase M | Usually the minus sign (-) in a differential pair |
| Pound sign (#), or overbar ($\overline{\text{bar}}$) | Lowercase or uppercase N | The overbar or pound sign (#) is often used for negative or active low pins |

> **Note**
>
> Some exceptions for special characters apply to the Pin Mate and Pin Electrical Characteristics columns in the model template. Mathematical operators such as the asterisk (*) are valid in column F (Pin Mate), and the asterisk (*), slash (/), plus sign (+), and minus sign (-) are valid in columns K to T (Pin Electrical Characteristics). Special characters are also allowed in the model description.

> **Note**
>
> Template cells A47 and B47 contain comments about invalid characters that are not recommended when editing #Pin Number or Pin Name columns.

## The Bulk Component Model Template

You can use a single spreadsheet to specify a list of bulk components, such as capacitors or resistors, rather than use an Intelligent Model Template for each part.

The following table provides the values column headers required for each type of bulk component. Values in red are required, values in blue are optional.

| 1 | | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | **Capacitors:** | #Cust Port No | MFG Part No | Description | Value | Tol | Volt | Class | Max Temp | Min Temp |
| 3 | **Inductors:** | #Cust Port No | MFG Part No | Description | Value (H) | Tol | I(mA) | Class | Max Temp | Min Temp |
| 4 | **Ferrite Beads:** | #Cust Port No | MFG Part No | Description | Value (Ohm) | Tol | I(mA) | Class | Max Temp | Min Temp |
| 5 | **Resistors:** | #Cust Port No | MFG Part No | Description | Value | Tol | Pwr(mW) | Max Temp | Min Temp | |
| 6 | **Fuse:** | #Cust Port No | MFG Part No | Description | V Rating (DC) | I(mA) | Max Temp | Min Temp | | |
| 7 | **Diodes:** | #Cust Port No | MFG Part No | Description | Fwd Voltage | Fwd Current (mA) | Reverse Voltage | Class | Max Temp | Min Temp |

For each type of component, you populate the column headers A through I in a spreadsheet. Then, on each subsequent row, you enter the data for each component. For example, a bulk component spreadsheet for capacitors could look like the following:

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | #Cust Part No | MFG Part No | Description | Value | Tol | Volt | Class | Max Temp | Min Temp |
| 2 | 1276-6456-2-ND | CL21A106KPFNNNG | Cap CER 10uf x5R 0805 | 10u | 5 | 10 | CER | 125 | -55 |
| 3 | 490-3160-1-ND | GRM0335C1E101JA01D | Cap CER 100pf 25V C0G/NP0201 | 100p | 5 | 25 | CER | 125 | -55 |

> **Note**
>
> During bulk import of capacitors and diodes, the tool sets the polarity using pin types. For polarized capacitors (non-CER and non-METP), pin 1 is type CP and pin 2 is pin type CM. For diodes, pin 1 is pin type DK and pin 2 is pin type DA.

> **Note**
>
> For TVS diodes, use TVS class for unidirectional TVS to auto-assign DA/DK pin type, and TVSB class for bidirectional TVS to auto-assign D pin type during import.

After you save the spreadsheet, you can import it to the model library, and make further edits to the component specifications. See **Manipulating the Contents of the Model Library**.

### Related Topics

**Manipulating the Contents of the Model Library**

## Completing the Model

After entering the component data into a model template, prepare the model for importing into the project's model library.

### Prerequisites

- You have entered the component data from the datasheet into a model template. See **HyperLynx Schematic Analysis Intelligent Model Template**.
- The model spreadsheet is open in the Microsoft Excel or Open Office tool.

### Procedure

1. In the model, confirm that the Flag cell at the top right corner of the template states "OK".

   This is a simple pin count check to see if there are an equal or greater number of pins modeled in the Pin Data columns as expected by the Pin Count cell.

   If the ListCount cell counts less pins, the cell is emphasized in red, and the Flag cell states the negative equivalent of the number of pins missing.
2. Save the completed model template in your project folder.

   A best-practice is to use the manufacturing number of the device as the name of the file, or a similar differentiator.

### Results

The model is ready to import into the model library. Refer to **Manipulating the Contents of the Model Library** for more information on how to import the model into the model library.

# 2. Model Template Contents

This chapter describes the contents of the model template, and where to locate the required information from datasheets.

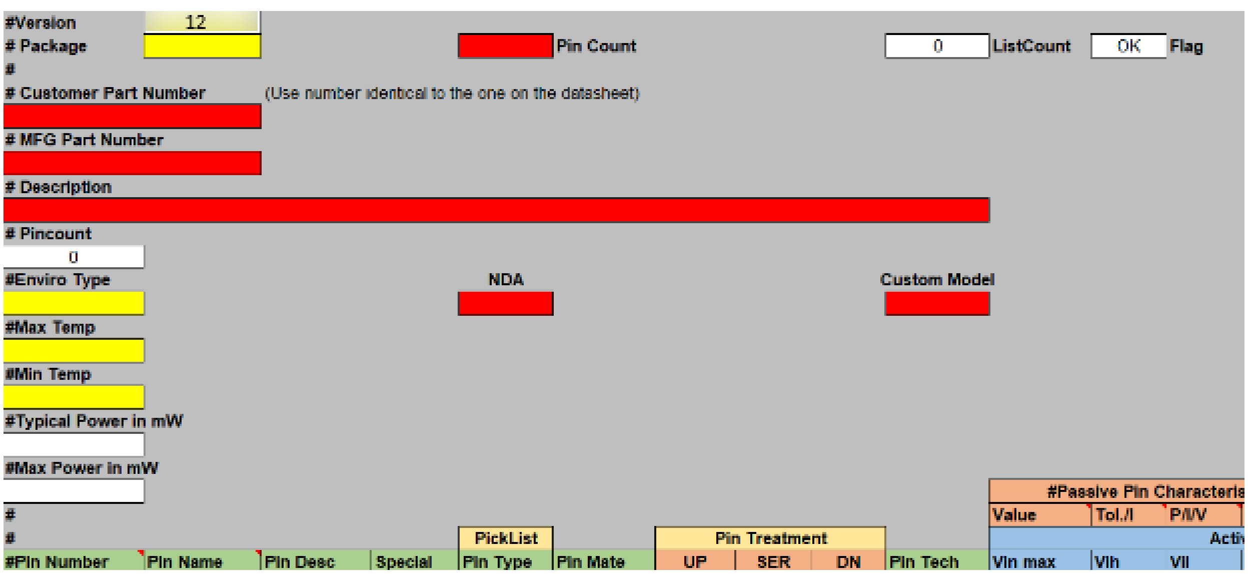


## Model Header

The model header describes the generic information of a component, such as its manufacturer part number and the number of pins on the device.

Figure 2-1 displays the fields in the model header. The required fields are emphasized in red, and the other fields are optional. However, if the datasheet provides information for the optional fields, you should enter that information.

**Figure 2-1: The Entry Fields of the Active and Passive Model Template Header**




When you create a model, much of the information you need is in the Ordering Information section of the related datasheet. To find fields in this section (such as package, number of pins, operating temperature, and environmental type), search for the Manufacturer Part Number in the datasheet. You can also use a part numbering guide to decode the Manufacturer Part Number.

## Model Header Cells

The following table lists the cells in the model header, and describes the information you enter into those cells.

**Table 2-1: Model Header Cells**

| Cell Name | Description |
|---|---|
| **Customer Part Number**<br><br>This cell is required | The Customer Part Number (CPN) is the designator name that a company assigns to a unique component in their BOM for their own database of electrical components. If a component is not assigned an individual CPN from a company's Approved Vendor List (AVL), then assign the CPN the same designator name as the manufacturer's part number. |
| **Manufacturer Part Number**<br><br>This cell is required | The manufacturer of the component provides the manufacturer part number, which you can find in the project's BOM. The part number is unique to the component, but might have similar versions depending on the select features of the device, such as supply voltage and frequency.<br><br>**Caution**<br>Model the component based on its manufacturer part number as specified in the BOM. The tool can then emphasize any possible unsynchronized errors between the BOM, netlist, or schematic files during the analysis. |
| **Package**<br><br>This cell is optional | Although optional, entering the package information is highly recommended to identify the correct package of the specified device. You can find which package to use for the device in the Ordering Information table or on the first few descriptive pages of the datasheet. Because there can be different versions of a component, ensure that the specified package is correct by comparing the MPN from the BOM and matching it with the list of part numbers in the Ordering Information table. |
| **Pin Count**<br><br>This cell is required | Enter the total number of pins of the component into cell E25. This information is dependent on the specific package and can normally be found in the Ordering Information on the component datasheet.<br><br>**Note**<br>The tool counts Exposed pads (EPADs) as pins. The tool reflects this information in cell J25 (List count), and you should update that pin in the Pin Count cell as well. |

**Table 2-1: Model Header Cells (continued)**

| Cell Name | Description |
|---|---|
| **Description**<br><br>This cell is required | You can copy the description of the device from the BOM description, or from the title of the device in the component's datasheet. The description should be clear enough so that another person reviewing or selecting models can easily read and understand the purpose of the device. It may also be useful to include specific characteristics such as memory size, configuration, and optional pins to distinguish between different versions of the device. |
| **Environmental Type**<br><br>This cell is optional | The device datasheet might mention if the component meets Restriction of Hazardous Substances (RoHS) standards. The Ordering Information table typically shows this, or it may be on the first descriptive page of the datasheet. What you enter in the cell depends on the information provided, as follows:<br><br>• No Lead: Enter 5/6 in cell A36 for components that are specified as only RoHS, Lead-Free, Pb-Free, or No PB.<br>• No Lead: Enter 6/6 in cell A36 for components that are specified as RoHS Green, PB-Free packaging.<br>• No Information: Leave this cell blank if there is no information on the datasheet regarding Environment Type. |
| **Operating Temperature (°C)**<br><br>This cell is optional | Enter the recommended Operating Temperature range (maximum and minimum) of the device in degrees Celsius. Use the ambient temperature or junction temperature if you do not know the operating temperature. The Ordering Information table often specifies the operating temperature of the specific device. If not, you can find it in the Recommended Operating Conditions or the DC Electrical Characteristics sections of the datasheet. |
| **Power Dissipation (mW)**<br><br>This cell is optional | Enter the average and/or maximum Power Dissipation of the component in milliwatts, if the datasheet provides the information. Usually, you can find this in the Absolute Maximum Ratings of the datasheet. |
| **NDA**<br><br>This cell is required | If the information in the device model requires a Non-Disclosure Agreement (NDA), you must enter YES in the NDA field; otherwise, enter NO. |

**Table 2-1: Model Header Cells (continued)**

| Cell Name | Description |
|---|---|
| | **Note**<br>A programmable device that is specific to a project, such as an FPGA, is automatically an NDA device. |
| **Custom Model**<br><br>This cell is required | If the modeled device varies from the datasheet or is specific to that model because of its functionality, you must enter YES in the Custom cell; otherwise, enter NO. A custom model generally refers to a specific operation of a device for a project or when a device is programmable and specific to the project. A model is also considered as custom when the datasheet is not available and the component must be modeled based only on the schematic.<br><br>Figure 2-2 shows an example of a model that has custom pins. This device is considered to be custom because its differential output pins can have different thresholds based on the pin technology. |

**Figure 2-2: Model Example of Custom Model Pins**

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 45 | # | | | | | | | | | | Value | Tol./I | P/W | Class | | | | | | |
| 46 | # | | | | PickList | | Pin Treatment | | | | Active Pin Characteristics | | | | | | | | For Power Pins | |
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | UP | SER | DN | Pin Tech | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 48 | 2 | CLKOUTp | | | DOUTP | 3 | | | | LVDS | | | | | VDDO | 2 | 0.6 | 0 | | |
| 49 | 3 | CLKOUTm | | | DOUTM | 2 | | | | LVDS | | | | | VDDO | 2 | 0.6 | 0 | | |
| 50 | #2 | CLKOUTp | | | DOUTP | 3 | | | | LVPECL | | | | | VDDO | 1.9 | 0.2 | 0 | | |
| 51 | #3 | CLKOUTm | | | DOUTM | 2 | | | | LVPECL | | | | | VDDO | 1.9 | 0.2 | 0 | | |
| 52 | 6 | GND | | | GND | | | | | | | | | | | | | | 0 | |
| 53 | 1 | EN | | | IN | | | | | | VDDO | 2 | 0.8 | 0 | | | | | | |
| 54 | 5 | IN | | | IN | | | | | | VDDO | 2 | 0.8 | 0 | | | | | | |
| 55 | 4 | VDDO | | | PWR | | | | | | | | | | | | | | 3.6 | 3 |

**Tip**

The HyperLynx Schematic Analysis tool imports only the rows that do not have a pound sign (#) under the #PinNumber column for that row. You can use the pound sign to comment out a row by adding a # at the beginning of the line. Therefore, depending on how the device is set up, if the chosen pin tech of the device is unused, add the # before the pin number of the unused pin, and leave the used pins as normal.

## Pin Data: Non-Electrical Information

The pin data columns describe two main elements of the device: the non-electrical and electrical information for the pin. This section explains the non-electrical pin information, such as the Pin Type and Pin Mate category.

The figure refers to which non-electrical pin data columns are available in the template.

**Figure 2-3: Non-Electrical Pin Data Fields of the Intelligent Model Template**

| | A | B | C | D | E | F | G | H | I | J | U |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 46 | # | | | | PickList | | Pin Treatment | | | | |
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | UP | SER | DN | Pin Tech | Pin Comment |

**Caution**

You must replace invalid symbols from the non-electrical section of the Pin Data columns. See **Table 1-2**. You can use the Check Model macro to automatically replace invalid symbols.

For information about the electrical pin data columns available in the intelligent model template, refer to **Pin Data: Electrical Information**.



## Pin Number and Pin Name

You must provide the Pin Number and the corresponding Pin Name for each pin according to the device package, with one pin or pad per row in the model template. The Pin Number and Pin Name are both required.

The cells in the Pin Number column allow any alphanumeric character. The value is typically a number, except in the following cases:

- **BGA** — The Ball Grid Array package pins are usually a combination of letters and numbers, such as AA00, BC12, or C28.
- **PAD/TAB** — Some devices have an exposed pad or case tab labeled EPAD, PAD, TAB, and so on. If the package of the device has such a pad, the tool models it as an additional pin number, and you should update that pin in the Pin Count cell. You can specify alternate pin names using the PIPE ( | ) character to separate the names, such as EPAD|49.
- **BJT/FET** — Some datasheets indicate Bipolar Junction Transistor pins as B, C, and E and Field Effect Transistor pins as G, D, and S. If a datasheet leaves this pin blank, refer to the netlist for the correct value.
- **DIODE** — Some datasheets indicate diode pins as A, K, or other similar combinations. If a datasheet does not specify the pin number it can be obtained from the symbol or netlist. You can specify several

pin number options using the PIPE ( | ) character to separate the pin numbers, such as 1|C|K for the cathode, and 2|A for the anode.

- **VREF** — The Pin Number column must be left blank if the device requires or has an internal reference voltage that does not have its own pin. See **Power and Ground Pins** for more details.

When referring to the pinout connection diagram on the datasheet, be sure to reference the correct package.

**Caution**

The pin numbers on the intelligent model must match the pins in your netlist. If your netlist has pin numbers that differ from the datasheet, then you have to modify the model or your netlist to make the model pin number match the netlist pin number.

You can use the **Run Data Confirmation** button in the HyperLynx Schematic Analysis **Project** tab to identify any unmodeled pins.

## Pin Description and Special Columns

You can use a cell in the Pin Description column to describe the pin. You can use a cell in the Special column for anything. The cells in the Pin Description column and Special column are optional.

### Pin Description Column

You can use cells in this column to describe the pin. You can also use this column as a sorting tool to segregate the different types of pins by adding a category name. For example, you could label this column as “JTAG” or “I2C” for associated JTAG or I2C pins. This column is especially useful for categorizing of pins for larger complex devices.

### Special Column

The Special column allows for free text, and you can use it to categorize pins for ease in model creation. For example, this column can list the dedicated power banks of an FPGA by entering the associated bank number (for example, 0 or 1 for Bank 0 or Bank 1, respectively). This column is useful when using equations in your spreadsheet tool to change the name of power supply rails that differ by dedicated bank, such as VCCIO0 and VCCIO1.

## Other Pin Data Categories From the Datasheet

The remaining pin data categories rely on the rest of the details given in a datasheet.

The pin description table is a good place to start for information on each pin of the device. You should browse through the entire datasheet for more detail on how a pin works. For example, a block diagram or circuit application example may give a high-level indication of the pin function.

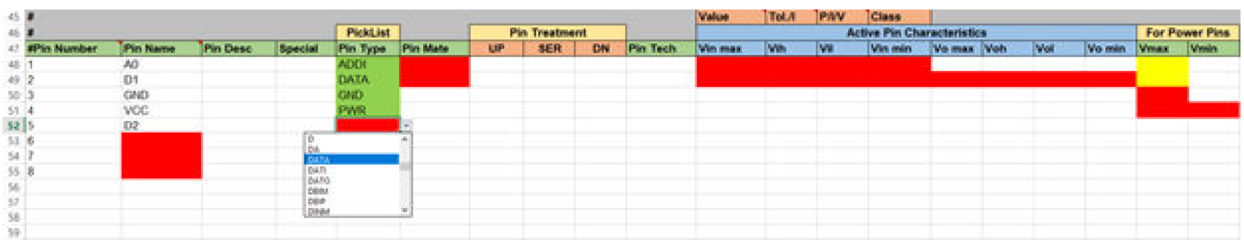
The Applications section also has more detail on certain pins. In more complex devices, such as micro-controllers or FPGAs, additional documents like Design Checklists or Hardware Design Guides might have more information on the device.

## Pin Type

The Pin Type refers to the nature of the pin (if it is an analog, digital, or power pin), its pin direction, and if it is single-ended or differential. This cell is required.

Figure 2-4 shows that each cell under the Pin Type column has a dropdown menu from which you can select the most appropriate type for each pin. After you select or enter a Pin Type, the pin data cells required for that Pin Type emphasize in red or yellow to indicate that you need to enter data in those cells.

**Figure 2-4: Available Pin Types in the Intelligent Model Template**



The following table lists all of the pin types available in the intelligent model template. In this table, the column labeled “Requires?” refers to the following:

- **Pin Mate** — This pin type requires a pin mate. See Pin Mate for more information on pin mates.
- ***Pin Mate** — This pin type does not require a pin mate, but it may be required depending on the device (if the device is a buffer or level translator).
- ***Pull-Up** — This pin type requires an external pull-up resistor. However, the HyperLynx Schematic Analysis tool does not require you to model an external pull-up resistor, because these pin types automatically account for them during an analysis.
- **PWROUT and GNDOUT** — These pin types are used for devices with internal regulators that supply power to external devices. If the device requires a PWROUT pin, you must redefine one of the GND pins in that device as a GNDOUT pin. GNDOUT signifies a ground source on the net where the pin is located.

**Table 2-2: The Pin Types Available in Intelligent Model Template**

| Pin Type | Pin Description | Pin Direction | Pin Category | Requires? |
|---|---|---|---|---|
| ADDB | Address bus | Bidirectional | Digital | Bus Significance |
| ADDI | Address bus | Input | Digital | Bus Significance |
| ADDO | Address bus | Output | Digital | Bus Significance |
| AIN | Analog | Input | Analog | - |
| AN | Analog | Bidirectional | Analog | *Pin Mate for level translators |
| AOUT | Analog | Output | Analog | - |
| BJTB | Bi-Polar Junction Transistor Base | Input | Digital | Pin Mate to BJTE |
| BJTC | Bi-Polar Junction Transistor Collector | - | Analog | Pin Mate to BJTE |
| BJTE | Bi-Polar Junction Transistor Emitter | - | Analog | - |
| C | Capacitor non-polarized | - | Passive | Pin Mate |
| CM | Capacitor polarized negative pin | - | Passive | Pin Mate |
| CP | Capacitor polarized positive pin | - | Passive | Pin Mate |
| DATA | Data bus | Bidirectional | Digital | Bus Significance |
| DATI | Data bus | Input | Digital | Bus Significance |
| DATO | Data bus | Output | Digital | Bus Significance |
| AINM | Analog Input Negative pin | Input | Analog | Pin Mate to AOUT, OUT or ODOUT pin types for op-amp or comparator feedback loop |
| AINP | Analog Input Positive pin | Input | Analog | Pin Mate to AOUT, OUT or ODOUT pin types for op-amp or comparator feedback loop |
| D | Diode anode/cathode pin | Bidirectional | Passive | Pin Mate |
| DA | Diode anode pin | - | Passive | Pin Mate |
| DK | Diode cathode pin | - | Passive | Pin Mate |

**Table 2-2: The Pin Types Available in Intelligent Model Template (continued)**

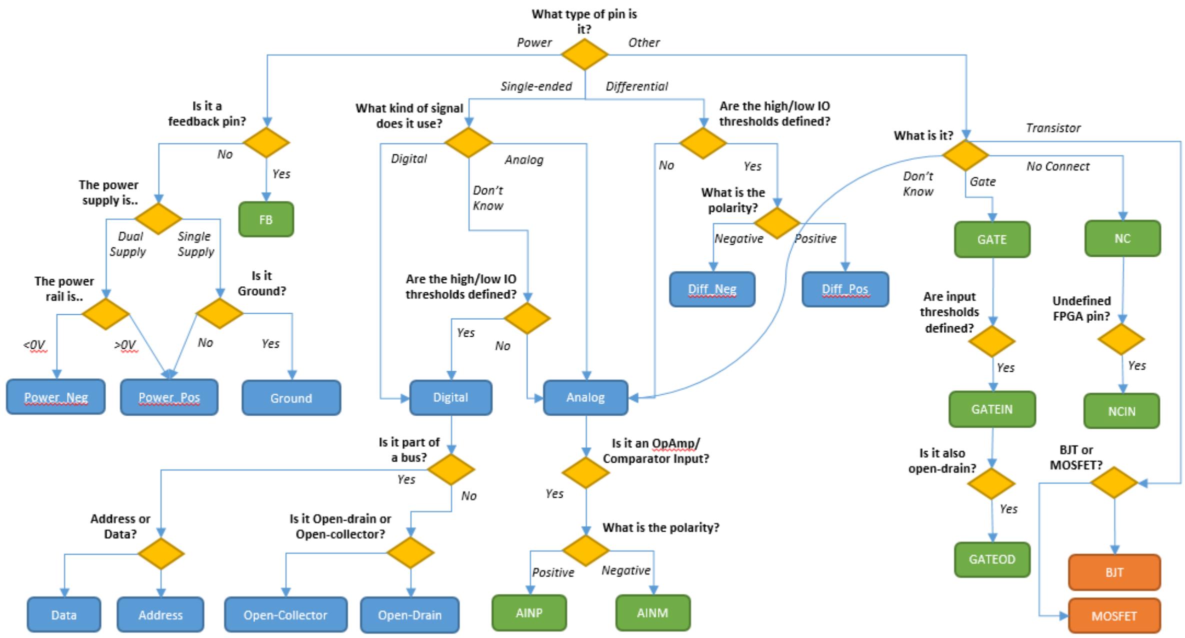
| Pin Type | Pin Description | Pin Direction | Pin Category | Requires? |
|---|---|---|---|---|
| DBIM | Differential negative | Bidirectional | Digital | Pin Mate |
| DBIP | Differential positive | Bidirectional | Digital | Pin Mate |
| DINM | Differential negative | Input | Digital | Pin Mate |
| DINP | Differential positive | Input | Digital | Pin Mate |
| DOUTM | Differential negative | Output | Digital | Pin Mate |
| DOUTP | Differential positive | Output | Digital | Pin Mate |
| BI | Digital | Bidirectional | Digital | *Pin Mate |
| IN | Digital | Input | Digital | *Pin Mate |
| OUT | Digital | Output | Digital | *Pin Mate |
| FETD | MOSFET drain | - | Analog | Pin mate to FETS |
| FETG | MOSFET gate | Input | Digital | Pin mate to FETS |
| FETS | MOSFET source | - | Analog | - |
| GATE | Analog switch | Bidirectional | Analog | Pin Mate |
| GATEIN | GATE or digital input | Input | Digital | Pin Mate |
| GATEOD | GATE or digital open-drain | Bidirectional | Digital | Pin Mate |
| NC | No connect | - | - | - |
| OC | Digital open-collector | Bidirectional | Digital | *Pull-Up |
| OCOUT | Digital open-collector | Output | Digital | *Pull-Up |
| OD | Digital open-drain | Bidirectional | Digital | *Pull-Up |
| ODOUT | Digital open-drain | Output | Digital | *Pull-Up |
| FB | Reference for feedback voltage | - | Power | - |
| GND | Ground | - | Ground | - |

**Table 2-2: The Pin Types Available in Intelligent Model Template (continued)**

| Pin Type | Pin Description | Pin Direction | Pin Category | Requires? |
|---|---|---|---|---|
| GNDOUT | Ground for PWROUT | - | Ground | PWROUT |
| PWR | Power supply to device | Input | Power | - |
| PWROUT | Power supply from device | Output | Power | GNDOUT |
| PWRN | Negative voltage supply pin of a dual supply device | Input | Power | Pin Mate to PWR pin |
| FUSE | Passive only - fuse | - | Passive | Pin Mate |
| I | Passive only - inductor or ferrite bead | - | Passive | Pin Mate |
| R | Passive only - resistor | - | Passive | Pin Mate |
| J | Passive only - connector pin | - | - | - |
| UNDEF | Undefined programmable pin | Bidirectional | Digital | - |
| XTAL | Crystal, for crystal oscillator devices only. Not to be used for crystal/clock input pins on active devices. | - | - | Pin Mate |
| CTAP | Transformer center tap | - | - | Pin Mate to XFMR pin |
| XFMR | Transformer multi-tap pin | - | - | Pin Mate |

You can use the decision tree in **Figure 2-5** to help you select the correct pin type based on the pin definition in the manufacturer datasheet.

**Figure 2-5: Pin Type Decision Tree**




| Direction | Digital | Address | Data | Open-drain | Open-Collector | Analog | Diff_Pos | Diff_Neg | Power_Pos | Power_Neg | Ground |
|---|---|---|---|---|---|---|---|---|---|---|---|
| INPUT | IN | ADDI | DATI | | | AIN | DINP | DINM | PWR | PWRN | GND |
| OUTPUT | OUT | ADDO | DATO | ODOUT | OCOUT | AOUT | DOUTP | DOUTM | PWROUT | PWROUT | GNDOUT |
| BI-DIRECTIONAL | BI | ADDB | DATA | OD | OC | AN | DBIP | DBIM | | | |

## Pin Mate

The cell in the Pin Mate column is required if the pin mate cell turns red after you enter a pin type value. There are different categories of pin mate depending on the selected pin type: it can identify the location of a bus within the device or indicate the relationship between the positive and negative mate pins of a differential pair. The pin mate can also indicate how the device affects other devices, such as whether it drives other devices as a bus master, or if the device pin is buffered or a gate.

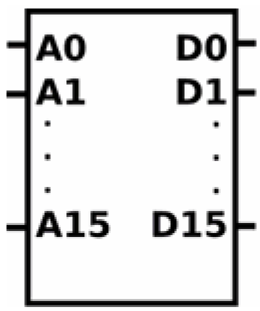
The following topics provide the information required in the pin mate cell for pin types. For more information on pin types, see Pin Type.

## Address and Data Bus

You must provide a pin mate value when the pin type is ADDB, ADDI, ADDO, DATA, DATI, or DATO. The address and data bus category of pin mate defines the size of the individual bus and the specific pin's location on the bus.

The syntax for this expression is LOCATION_SIZE. The bus format can be either little endian or big endian. The most common format is little endian, where a 16-bit data bus has a least significant bit of 0 16 and a most significant bit of 15 16. The significant bits for the big endian format are reversed. Figure 2-6 shows an example of how you label the Pin Mate column using little endian or big endian format.

**Figure 2-6: Model Example of the Bus Pin Mate Using Little Endian and Big Endian Formats**

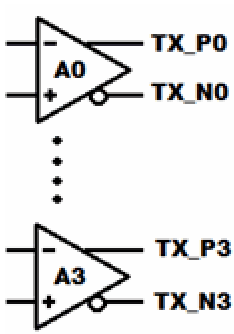



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 48 | A1 | ADDR_A[0] | LittleEnd_0 of 16 | | ADDI | 0_16 |
| 49 | A2 | ADDR_A[1] | LittleEnd_1 of 16 | | ADDI | 1_16 |
| 50 | A16 | ADDR_A[15] | LittleEnd_15 of 16 | | ADDI | 15_16 |
| 51 | | | | | | |
| 52 | B1 | ADDR_B[0] | BigEnd_15 of 16 | | ADDI | 15_16 |
| 53 | B2 | ADDR_B[1] | BigEnd_14 of 16 | | ADDI | 14_16 |
| 54 | B16 | ADDR_B[15] | BigEnd_0 of 16 | | ADDI | 0_16 |

The Pin Mate column also supports bit swizzling of data buses (different ordering on receive and transmit). To enable bit swizzling, in the Pin Mate cell for the pin, append a pipe "|", followed by an "S" and the number of bits to swizzle. For example, for a 16-bit bus width, without swizzling enabled, the first bit's Pin Mate cell may read "$0_16" - with swizzling the entire bus enabled it would read "$0_16|S16", with swizzling only the first 8 bits enabled it would read "$0_16|S8".

The address and data bus pin mate syntax can also apply to other pins that have similar bus functionality. In a complex device, you indicate the bus functionality by multiple instances of the same pin name with brackets surrounding the location of the pin.

**Figure 2-7: Model Example of the Bus Pin Mate With Differential Transmitter Output Pins**



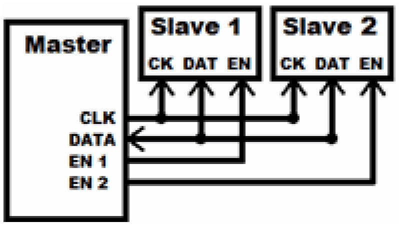

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 56 | W1 | TX_N[0] | OUT OF 4 | | DOUTM | N2\|$0_4 |
| 57 | U1 | TX_N[1] | OUT OF 4 | | DOUTM | Y2\|$1_4 |
| 58 | E1 | TX_N[2] | OUT OF 4 | | DOUTM | F2\|$2_4 |
| 59 | T1 | TX_N[3] | OUT OF 4 | | DOUTM | S2\|$3_4 |
| 60 | N2 | TX_P[0] | OUT OF 4 | | DOUTP | W1\|$0_4 |
| 61 | Y2 | TX_P[1] | OUT OF 4 | | DOUTP | U1\|$1_4 |
| 62 | F2 | TX_P[2] | OUT OF 4 | | DOUTP | E1\|$2_4 |
| 63 | S2 | TX_P[3] | OUT OF 4 | | DOUTP | T1\|$3_4 |

## Bus Master

A pin mate might be required when the pin type is ADDB, ADDO, DATA, or DATO pin. The bus master category of pin mate typically has a bus with a designated bus master and multiple slave devices attached.

The slave device can be a memory device, and the master device is the programmable chip that reads and writes to and from memory. A dollar sign ($) before the LOCATION_SIZE syntax in the Pin Mate column signifies that the driver in this device is the Bus Master.

**Figure 2-8: Model Example of the Bus Master Pin Mate**



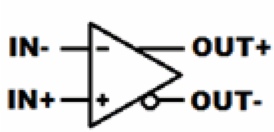

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 69 | G1 | DATA0 | OUT OF 16 | | DATO | $0_16 |
| 70 | G2 | DATA1 | OUT OF 16 | | DATO | $1_16 |
| 71 | G16 | DATA15 | OUT OF 16 | | DATO | $15_16 |

## Differential Pairs

You must provide a pin mate value when the pin type is DBIM, DBIP, DINM, DINP, DOUTM, or DOUTP. The differential pair category of pin mate must state the relationship between the positive and negative pins of an IC in the pin mate cell.

Enter the pin number of the differential pin's complement into the pin mate cell.

**Figure 2-9: Model Example of the Differential Pin Mate**




| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 56 | E12 | CLKINp | | | DINP | D12 |
| 57 | D12 | CLKINm | | | DINM | E12 |
| 58 | M3 | CLKOUT0p | OUT OF 4 | | DOUTP | N3\|$0_4 |
| 59 | N3 | CLKOUT0m | OUT OF 4 | | DOUTM | M3\|$0_4 |

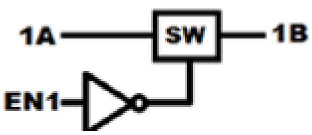
> **Tip**
>
> Cells in the pin mate column can have a combination of these pin mate types by separating the information in the cell with a pipe ( | ).

## Gates and Switches

You must specify a pin mate when the pin type is GATE, GATEIN, or GATEOD. For gates and switches, the cell in the pin mate column must state the relationship between the pins on either side of the component in the pin mate cell.

You enter the pin number of the devices complement into the pin mate cell.

**Figure 2-10: Model Example of the Gate and Switches Pin Mate**



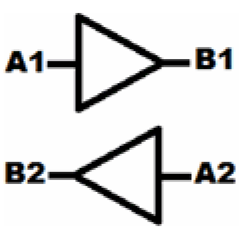

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 48 | 1 | SWITCH_1A | | | GATE | 2 |
| 49 | 2 | SWITCH_1B | | | GATE | 1 |

## Buffer and Level Translators

A pin mate might be required if the pin type is BI, IN, or OUT. The pin mate for buffers and level translators is similar to the pin mate for gates and switches.

You enter the pin number of the device's complement into the pin mate cell. In addition, an asterisk (*) must precede the complementary pin number. This allows the HyperLynx Schematic Analysis tool to preserve the bus significance on either side of the device.

**Figure 2-11: Buffer and Level Translators Pin Mate**



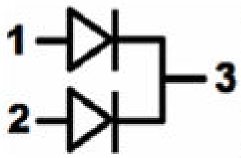

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
| 64 | 3 | SIDE_A1 | | | IN | *4 |
| 65 | 4 | SIDE_B1 | | | OUT | *3 |
| 66 | 5 | SIDE_A2 | | | IN | *6 |
| 67 | 6 | SIDE_B2 | | | OUT | *5 |

## Passive Components

In the intelligent model template, you must specify the relationship of the pins on either side of passive components (such as diodes, resistors, inductors, and capacitors) in the Pin Mate cell.

You enter the pin number of the passive component's complement into the Pin Mate cell. See the following figure for an example of a device with two diodes tied together by their cathode pin.

**Figure 2-12: Passive Component Pin Mate**




| 47 | #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate |
|---|---|---|---|---|---|---|
| 48 | 1 | ANODE1 | | | DA | 3 |
| 49 | 2 | ANODE2 | | | DA | 3 |
| 50 | 3 | CATHODE | | | DK | 1 |
| 51 | 3 | CATHODE | | | DK | 2 |

For passive components with more than two pins, such as resistor arrays, you must make multiple rows of the same pin to reflect the pin mate relationship between each component in the array. Therefore, in the preceding figure, two rows were made for the cathode pin to describe its connection with anode1 and anode2.

## Conditional Pin Function

Use the Pin Mate column for conditional pin function situations. One conditional pin function situation could be a pin number that has multiple pin names and associated functions, and the operating pin name and function depend on the state of another pin number.

Populate the dependent Pin Mate cell with the control pin's name followed by the control pin's state in parentheses.

> **Note**
>
> The control pin must be a digital input pin with a defined threshold, and can be driven as long as the pin is pulled to a known state.

You set the control pin's state to a level using pullups and/or pulldowns as a voltage divider.

- LOW is a level below the $V_{IL}$ threshold.
- HIGH is a level above the $V_{IH}$ threshold.
- MID is a level between $V_{IL}$ and $V_{IH}$.

If the selector pin is driven with no pullups or pulldowns, the tool cannot determine the selector level and generates an error stating that it cannot determine which function is being selected. If the selector pin is driven with a pullup, pulldown, or voltage divider, the HyperLynx Schematic Analysis tool defaults to the state defined by the pullup or pulldown.

> **Note**
>
> In the example shown in **Figure 2-13**, depending on which signal pin 3 (Pin Name=MODE) receives (logic HIGH, MID, or LOW), the device output at pins 1 and 2 will be single-ended CMOS output, or differential LVDS output, or LVPECL output pin tech, each of which would have different thresholds.

> **Note**
>
> For differential pin technologies, such as LVDS or LVPECL, you need to define the N-side (input pin) or P-side (clock pin) of a differential pair in the Pin Mate column. You use a pipe character (|) to indicate an AND operation between the differential pin and the control pin as shown in **Figure 2-13**.

**Figure 2-13: Pin With Multiple Pin Functions**

| # | | | | PickList | | Pin Treatment | | | |
|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | UP | SER | DN | Pin Tech |
| 1 | OUTp | | | OUT | MODE(HIGH) | | | | CMOS |
| 2 | OUTm | | | OUT | MODE(HIGH) | | | | CMOS |
| 1 | OUTp | | | DOUTP | 2\|MODE(MID) | | | | LVDS |
| 2 | OUTm | | | DOUTM | 1\|MODE(MID) | | | | LVDS |
| 1 | OUTp | | | DOUTP | 2\|MODE(LOW) | | | | LVPECL |
| 2 | OUTm | | | DOUTM | 1\|MODE(LOW) | | | | LVPECL |
| 3 | MODE | | | IN | | | | | |

## Pin Treatments

HyperLynx Schematic Analysis pin treatments enable you to specify the circuit topology of a pin. You specify pin treatments using the columns UP, SER, and DN in the model template.

> **Note**
>
> The tool always applies the power-pin decoupling rules IDC and IDCB to power pins (pin type PWR, PWROUT, GND, GNDOUT, PWRN). However, the tool does not apply pin treatment values to power pins.

Pin treatment options allow a wide range of customization for models, depending on the design. For generic models, it is recommended to document only the mandatory pin treatment information. For example, a design might need a 100 µF capacitor to ground to set the frequency; this is not necessary for a generic model as the design may not always require a capacitor. Any custom-specific models must have the Custom cell marked “Yes”.

The pin treatment columns include:

- **UP** — A connection to a positive voltage rail
- **SER** — A connection in series with other pins
- **DN** — A connection to a negative voltage rail or GND

Use the pin treatment columns to model passive components with internal and external connections to the pin, such as resistors, capacitors, inductors, or diodes. You can also use these columns to model subjective pin treatments of the device, which refers to the following states: Active High or Low, Unused, No Connection, and Driven.

You can specify value ranges for resistors, capacitors, and so on in the pin treatments columns. The range must use the same postfix, such as R1K-10K.

The pin treatment columns support the unit definitions for resistors, capacitors, and inductors (# is a value) listed in Table 2-3.

**Table 2-3: Supported Unit Definitions**

| Resistors | Capacitors | Inductors |
|---|---|---|
| # (Ohms) | #U or #UF (Microfarads) | #MH (Millihenries) |
| #K (Kilohms) | #N or #NF (Nanofarads) | #UH (Microhenries) |
| #M (Megaohms) | #P or #PF (Picofarads) | #NH (Nanohenries) |

**Caution**

For resistors, lowercase "m" for "milli" is not supported.

Table 2-4 lists the available pin treatment options and the valid placement for each entry. Other options with units as per Table 2-3 are also supported. The "x" in the SER columns indicates a passive component connection to another device, and "x@#" in the SER column indicates a passive component connection from current pin to another pin on the same device.

**Table 2-4: Available Pin Treatment Options**

| Type | UP | SER | DN | Subjective Pin Treatment |
|---|---|---|---|---|
| A | X | | X | Enable the device to its normal mode of operation. |
| N | X | X | X | Treatment for unused pins, where UP/DN for direct connection to rail/GND, SER for unconnected. |
| X | X | X | X | Pin must not be pulled high/driven/pulled low. |
| Y | | X | | This pin must have a Driven Signal connection. |
| **Type** | **UP** | **SER** | **DN** | **Internal and External Passive Components** |
| - | | @# | | External direct connection to another pin on device. |
| N{value}K | X | | X | Resistor - External known resistor in kΩ for unused pin treatment. |
| NK | X | | X | Resistor - External unknown value in kΩ for unused pin treatment. |
| R | X | X,X@# | X | Resistor - External unknown value. |
| R{value} | X | X,X@# | X | Resistor - External known value in Ω. |

**Table 2-4: Available Pin Treatment Options (continued)**

| | | | | |
|---|---|---|---|---|
| R{value}K | X | X,X@# | X | Resistor - External known value in kΩ. |
| K | X | X,X@# | X | Resistor - Internal unknown value. |
| {value}K | X | X,X@# | X | Resistor - Internal known value in kΩ. |
| C | X | X,X@# | X | Capacitor - External unknown value. |
| C{value}U | X | X,X@# | X | Capacitor - External known value in μFarads. |
| U | X | X,X@# | X | Capacitor - Internal unknown value. |
| {value}U | X | X,X@# | X | Capacitor - Internal known value in μFarads. This type is deprecated: use {value}UF. |
| {value}UF | X | X,X@# | X | Capacitor - Internal known value in μFarads. |
| {value}N | X | X,X@# | X | Capacitor - Internal known value in nanoFarads. This type is deprecated: use {value}NF. |
| {value}NF | X | X,X@# | X | Capacitor - Internal known value in nanoFarads. |
| L | X | X,X@# | X | Inductor - External unknown value. |
| L{value}UH | X | X,X@# | X | Inductor - External known value in μHenries. |
| H | X | X,X@# | X | Inductor - Internal unknown value. |
| {value}UH | X | X,X@# | X | Inductor - Internal known value in μHenries. |
| {value}NH | X | X,X@# | X | Inductor - Internal known value in nanoHenries. |
| D, DA, DK | X | X,X@# | X | External diode. |
| P | | X | | Enables swapping of polarity on differential pairs. |
| **Type** | **UP** | **SER** | **DN** | **Additional Syntax** |
| “\|” (pipe) | X | X | X | Separates multiple pin treatment entries in a cell. |

**Table 2-5: Pin Treatment Examples**

| UP | SER | DN | Comment |
|---|---|---|---|
| A\|K | | | Pin is active high, enabling the operation of the device. Internal unknown value pullup resistor at pin. |
| N1K | | N1K | If pin is unused, it must be tied high or low by a 1kOhm resistor. Must not be left floating. |
| | C0.1U@14 | | Place a 0.1uF capacitor between this pin and pin 14. |
| R1K-10K | | | This pin must have a pullup resistor with a value range between 1kOhm and 10kOhm. |

## Pin Technology

The pin technology column in the intelligent model template is reserved for the name of standard voltage logic levels applied to a specific pin. The pin technology cell is optional for the model template. Although the pin technology cell is optional, it is best to specify the pin technology to help identify the logic level standard used for the voltage thresholds.

These logic levels should directly relate to the voltage thresholds acceptable for a specific pin on the device. Some examples of these common logic levels include single-ended thresholds such as CMOS, TTL, and SCHMITT trigger inputs, or differential thresholds such as LVPECL and LVDS.

## Pin Comments

You use pin comments to identify warnings, exceptions, or general comments that affect the specified pin. This cell is optional. These comments appear on the voltage threshold errors in the analysis.

> **Note**
>
> You can force output to the comment section of results by inserting test “FLAG:” prior to the comment text for that model’s pin. See **Intentional Errors**.

# Pin Data: Electrical Information

The pin data columns have a section for the DC electrical characteristics of the device. In this section, you explain the logic levels and other electrical characteristics and how they can be modeled in the template.

The figures show the relevant columns in the model templates.

**Figure 2-14: Electrical Pin Data Fields**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 43 | | | | | | | | | | |
| 44 | #Passive Pin Characteristics | | | | | | | | | |
| 45 | Value | Tol./I | P/I/V | Class | | | | | | |
| 46 | Active Pin Characteristics | | | | | | | | For Power Pins | |
| 47 | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 48 | | | | | | | | | | |
| 49 | | | | | | | | | | |

The device datasheet typically includes a DC Electrical Characteristics section or a Recommended Operating Conditions section. This is where the pin specifications for the device are typically located. Depending on the pin type and device, the electrical characteristics can be modeled differently for each one. This section explains the most common characteristics found in datasheets.

> **Note**
>
> The Absolute Maximum Rating section of a datasheet also lists specifications. However, you should avoid using data from this section because these ratings are at the extreme limits of the device and are not at the recommended voltages.



## Single-Ended Logic Levels

The HyperLynx Schematic Analysis tool measures single-ended signals and compares them to a voltage, typically ground.

In digital devices, there are only two logic states: on and off. These states are also called logic high for 1 or logic low for 0. For the receiver to acknowledge the signal as a logic high or low, the driver must ensure its signal is above the VIH threshold or below the VIL threshold, respectively. Figure 2-15 displays the logic thresholds as they apply for single-ended signals.

**Figure 2-15: Valid Output Voltage Range for a Receiver to Acknowledge a Logic High or Low**

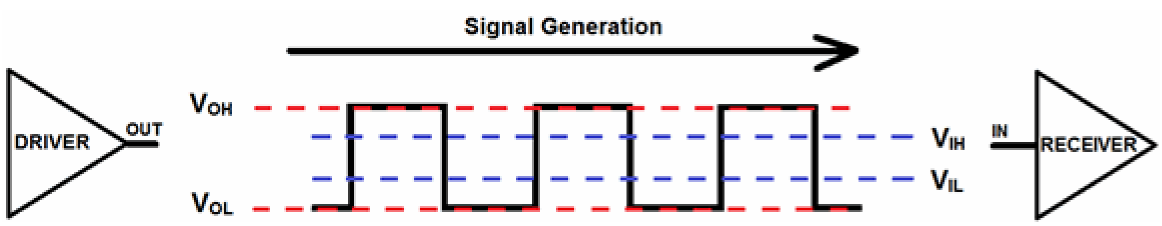



Table 2-6 explains the single-ended threshold levels as they apply to modeling in the HyperLynx Schematic Analysis tool.

**Table 2-6: Definitions for Single-Ended Logic Levels as Applied to the HyperLynx Schematic Analysis Tool**

| Single-Ended Input | Comment |
|---|---|
| VIN max | The maximum recommended voltage that can be applied to a specific pin. This can be the reference to the power supply, or the highest specified value found in the VI or VIH specifications. |

**Table 2-6: Definitions for Single-Ended Logic Levels as Applied to the HyperLynx Schematic Analysis Tool (continued)**

| | |
|---|---|
| | See the comment for VIN min in this table for more information when entering a numerical value into the cell instead of text. |
| VIH | The lowest value that the RECEIVER can detect for a valid input voltage high. |
| VIL | The highest value that the RECEIVER can detect for a valid input voltage low. |
| VIN min | The minimum recommended input voltage that can be applied to a specific pin. It is normally 0 volts, and the tool automatically provides the value. |
| **Single-Ended Output** | **Comment** |
| VO max | The maximum output voltage that a specific pin can supply. This can be the reference to the power supply or the highest specified value found in the VO or VOH specifications. |
| VOH | The lowest value that the pin can drive for a valid output voltage high. |
| VOL | The highest value that the pin can drive for a valid output voltage low. |
| VO min | The lowest output voltage that a specific pin can drive. It is normally 0 volts, and the tool automatically provides the value. |

When you select a Pin Type, either the VMAX cell, VMIN cell, or both emphasizes accordingly. The following applies to the VMAX and VMIN columns in this case:

- VMAX emphasizes in yellow when you enter a Pin Type as a reminder for the possibility of instantiations. See **Instantiations** for more information.
- VMIN emphasizes in red when VIN max is a numerical value and not a reference to a power supply, or it can be an instantiation that has been created through a value input to the VMAX cell. This requires the contents of cell VMIN to declare the power supply of the specific pin.

The recommended operating condition information for a device in a datasheet is typically straightforward when examining the logic levels for single-ended thresholds. Most datasheets provide the supply voltage rails, as well as the minimum and maximum high-level input voltage for A- and B-Port I/O pins. You can extract the $VIN_{max}$, VIH, VIL, and VIN min electrical characteristics from this information.

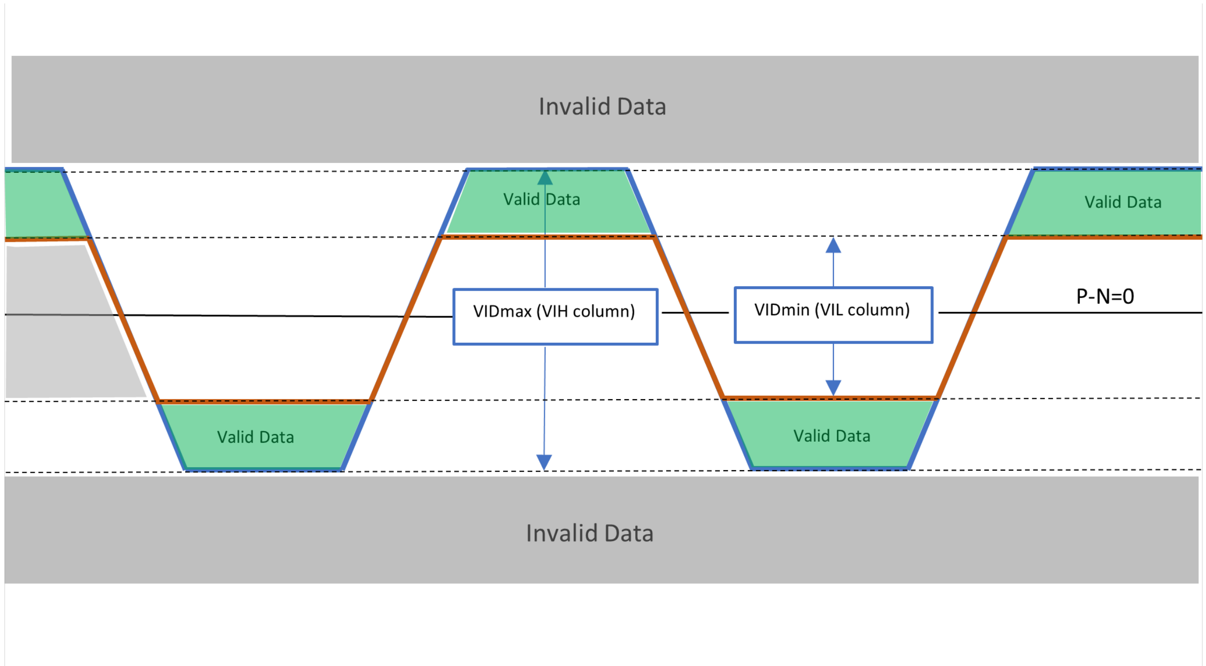
The model template enables you to insert equations in these cells. However, the syntax requires a space between the supply rail name and the mathematical operator and value. For example, "VDD[insert space here]*0.51", or written as "VDD *0.51" are valid in the template. The HyperLynx Schematic Analysis tool then calculates this value during the analysis.

## Differential Logic Levels

Differential-ended signaling takes the difference between two complementary signals rather than a reference to ground. For the receiver to register a valid signal, the driver output differential voltage must be below the maximum differential input voltage, VID max, and above the minimum threshold, VID min.

The green areas in the following figure represent where the driver signals are valid, and the gray areas represent the invalid regions.

**Figure 2-16: Valid Output Voltage Range for a Differential Receiver to Acknowledge a Valid Signal**




The table explains the differential thresholds as applied to modeling in the HyperLynx Schematic Analysis tool. In the active template, the pin data columns VID max, VID min, VOD max, and VOD min represent VIH, VIL, VOH, and VOL, respectively, for differential pins.

**Table 2-7: Definitions for Differential Thresholds Logic Levels as Applied to the HyperLynx Schematic Analysis Tool**

| Differential Input | Comment |
|---|---|
| VIN max | The maximum recommended DC input voltage that can be applied to a specific pin, if specified. This is typically the reference to the power supply. |
| VID max (VIH) | The maximum peak-to-peak differential input voltage.<br><br>(VIDmax = 2*VIDmax,pk) |

**Table 2-7: Definitions for Differential Thresholds Logic Levels as Applied to the HyperLynx Schematic Analysis Tool (continued)**

| | |
|---|---|
| VID min (VIL) | The minimum peak-to-peak differential input voltage.<br><br>(VIDmin = 2*VIDmin,pk) |
| VIN min | The minimum recommended DC input voltage that can be applied to a specific pin, typically 0 volts. |
| **Differential Output** | **Comments** |
| VO max | The maximum DC output voltage that can be output by a specific pin, if specified. This is typically the reference to the power supply. |
| VOD max (VOH) | The maximum peak-to-peak differential output voltage.<br><br>(VODmax = 2*VODmax,pk) |
| VOD min (VOL) | The minimum peak-to-peak differential output voltage.<br><br>(VODmin = 2*VODmin,pk) |
| VO min | The lowest DC output voltage that can be driven by a specific pin. It is normally 0 volts. |

The datasheet provides the recommended operating conditions of a device with differential input pins. The difficulty with differential thresholds is determining if the provided thresholds are in single-ended or differential-ended format. If the datasheet specifies a peak-to-peak ($V_{pkpk}$) voltage swing, this is typically the information needed, but not always. There might be more information provided in the datasheet, such as a graph of the voltage swing to show the definition of the characteristic, to confirm if it is differential.

The datasheet also provides the supply voltage rails and the differential peak-to-peak input swing of the device. You can extract the VID max (Column L) and VID min (Column M) electrical characteristics of a differential input pin from this information. Because there is no reference to other limitations, the maximum recommended DC input voltage, VIN max, will be referred to the voltage supply rail VDD. For VIN min, the value will be 0.0 V.

**Tip**

If a differential signal is specified as a peak voltage ($V_{pk}$), then double the value of the peak voltage to obtain the peak-to-peak value.

## Schmitt Triggers

Certain CMOS and TTL devices require Schmitt trigger inputs to operate.

Schmitt trigger specification input thresholds are labeled Positive-Going Input Threshold (VT+) and Negative-Going Input Threshold (VT-). When modeling a Schmitt trigger input, it is important to remember that the VIH is the maximum value for the VT+, and the VIL is the minimum value for the VT-.

## MOSFETs

The MOSFET is a 3-pin terminal device (not including the Body terminal) with a Gate, Drain, and Source terminal, modeled as FETG, FETD, and FETS, respectively.

The HyperLynx Schematic Analysis tool takes into account only the characteristics of the gate terminal. Table 2-8 explains the device thresholds as they apply to the input gate of the MOSFET.

**Table 2-8: Threshold Definitions for the MOSFET Input Gate Terminal**

| Pin Type | Pin Mate | $VIN_{MAX}$ | $V_{IH}$ | $V_{IL}$ | $VIN_{MIN}$ |
|---|---|---|---|---|---|
| FETG | FETS | Abs $Max_{VGS}$ | Lowest VGS @ $RDS_{(on)}$ | *Lowest* $VGS_{(th)}$ | 0 Volts |
| FETD | FETS | $VDS_{(max)}$ | | | |
| FETS | | | | | |

## Bipolar Junction Transistors

The Bipolar Junction Transistor (BJT) is a 3-pin terminal device with a Base, Collector, and Emitter.

**Table 2-9: Threshold Definitions for the BJT Terminals**

| Pin Type | Pin Mate | $VIN_{MAX}$ | $V_{IH}$ | $V_{IL}$ | $VIN_{MIN}$ |
|---|---|---|---|---|---|
| BJTC | BJTE | $VCE_{(max)}$ | | | 0 Volts |
| BJTB | BJTE | Abs Max VBE | *Highest* $VBE_{(sat)}$ | *Lowest* VBE(cut-off) | 0 Volts |
| BJTE | | | | | 0 Volts |

## Open-Drain/Collector Outputs

An open-drain or open-collector output pin is a transistor (either a BJT or a MOSFET) with its drain or collector pin left floating (also known as open or high-impedance) and the source or emitter tied to ground.

A driver can turn the transistor on to pull the signal to ground or leave it open when the transistor is off. When the transistor is off, the signal can be pulled up or down by an external resistor to prevent an undefined, floating state.

When modeling the output characteristics of an open-drain pin in the HyperLynx Schematic Analysis tool, the VOL is the highest value that the pin can drive a valid output voltage low. VOL is usually provided in the Recommended Operating Conditions of the datasheet. Because the output is externally pulled up high by a resistor, the VO max and VOH values are typically the power supply rail. Some datasheets specify a VOH value for open-drain pins.

## GNDOUT Versus GND Pins

The decision to use a GNDOUT or GND pin type depends on the device you are modeling.

- For voltage regulators, at least one GND pin should be modeled as GNDOUT pin type.
- For all other devices use the GND pin type for GND

## Passive Component Characteristics

You specify electrical characteristics and pin technology for passive components.

Table 2-10 summarizes electrical characteristics of passive components. The empty cells in the table indicate that the model needs no information in that cell.

**Table 2-10: DC Characteristics Required for Each Passive Component**

| Component | Pin Type | Pin Mate | Value | Tolerance | P/I/V | Class |
|---|---|---|---|---|---|---|
| Diode | D, DA, DK | Required | See Table 2-12 | See Table 2-12 | See Table 2-12 | See Table 2-12 |
| Capacitor | C, CP, CM | Required | Farads[1] | % | Volts | See Table 2-11 |
| Resistor | R | Required | Ohms[2] | % | mWatts | |
| Inductor | I | Required | Henries[3] | % | mAmps | IND |

1 Capacitors accept multipliers of U (micro), N (nano), and P (pico).
2 Resistors accept multipliers of K (kilo), M (mega), or no multiplier for Ohms. Lowercase m for milli, is not supported.
3 Inductors accept multipliers of M (milli), U micro), and N (nano).

**Table 2-10: DC Characteristics Required for Each Passive Component (continued)**

| Componen t | Pin Type | Pin Mate | Value | Tolerance | P/I/V | Class |
|---|---|---|---|---|---|---|
| Ferrite Bead | I | Required | Zo @ 100 MHz | % | mAmps | BEAD |
| Fuse | FUSE | Required | $V_{rating}$ DC | | mAmps | |

**Caution**

For resistors, lowercase "m" for "milli" is not supported.

After you specify a pin type, you add the component class (type) to the Class column of the intelligent model template for capacitors, diodes, and inductors. Table 2-11 displays the available classes in the dropdown menus of the Class cell on the intelligent model template.

**Table 2-11: Available Capacitor Classes**

| Class | Type | Description | Voltage Type |
|---|---|---|---|
| AL | CM, CP | Aluminum or Aluminum Electrolytic | $V_{rating}$ DC (Volts) |
| ALO | CM, CP | Aluminum Organic | $V_{rating}$ DC (Volts) |
| CER | C | Ceramic / MLCC | $V_{rating}$ DC (Volts) |
| METP | C | Metalized Polyester / Plastic / Film | $V_{rating}$ DC (Volts) |
| TANTP | CM, CP | Polymer Tantalum | $V_{rating}$ DC (Volts) |
| TANTS | CM, CP | Solid Tantalum / MnO2 | $V_{rating}$ DC (Volts) |
| SCAP | Capacitor | Super Capacitor | $V_{rating}$ DC (Volts) |
| Note: The plus side of a polarized capacitor is pin 1 | | | |

You can find the Value, Tolerance, and Power/Current/Voltage of a passive component in the description of the component in the Bill of Materials. If the information is missing, normally a quick search online or in the datasheet can help identify the electrical characteristics of the component.

**Note**

The values you enter in these cells should not contain any units. However, you can enter numerical prefixes (such as pico, nano, and micro) in the cell after the value (for example, 10U for 10µF, or 50K for 50 kOhms).

## Diodes

When you select Pin Type D, the Pin Mate, P/I/V, and Class cells emphasize in red to indicate that these cells require an entry. Select the appropriate diode class and the respective pin mate for each pin. In the P/I/V column, enter the value of the recommended voltage (see **Table 2-12**) of the diode in volts. You can use the absolute maximum rating, if you cannot find the recommended value on the datasheet.

**Table 2-12: Recommended Voltage Types for Different Diode Pin Classes**

| Class | Pin Type | Pin Mate | Value | Tol. | P/I/V | Description |
|---|---|---|---|---|---|---|
| RECT | DA, DK | Required | $VF_{(max)}$ | $IF_{(max)}$ | $V_R$ | Rectifier |
| SCHOTTKY | DA, DK | Required | $VF_{(max)}$ | $IF_{(max)}$ | $V_R$ | Schottky |
| SWITCH | DA, DK | Required | $VF_{(max)}$ | $IF_{(max)}$ | $V_R$ | Switching |
| LED | DA, DK | Required | $VF_{(typ)}$ | $IF_{(test)}$ | $V_R$ | Light Emitting Diode |
| ZENER | DA, DK | Required | $VF_{(typ)}$ | $I_{ZT}$ | $V_{Z(nom)}$ | Zener |
| TVS | D, DA, DK | Required | $V_R$[4] | $I_{PP}@V_C$ | $V_C$ | Transient Voltage Suppressor |

**Note**

Current is modeled in mAmps.

For 2 pin devices, Pin 1 is the Cathode.

## Capacitors

When you select the Pin Type C, the Pin Mate, Value, Tolerance, P/I/V, and Class, cells emphasize in red to indicate that these cells require you to enter a value. Select the appropriate pin technology and the respective pin mate. Enter the numerical value of the capacitor in Farads in the Value column. Enter the tolerance value of the capacitor in the Tolerance column. Enter the voltage rating of the capacitor in the P/I/V column in volts.

## Resistors

When you select the Pin Type R, the Pin Mate, Value, Tolerance, and P/I/V cells emphasize in red to indicate that these cells require you to enter a value. Select the respective pin mate for each pin. Enter the numerical value of the resistor in Ohms in the Value column, and the tolerance of the resistor in the Tolerance column. Enter the value of the power rating of the resistor in milliwatts in the P/I/V column.

4 $V_R$ (Reverse Voltage) is sometimes referred to as Stand-off voltage or Working Reverse Voltage.)

### Inductors

When you select the Pin Type I, the Pin Mate, Value, Tolerance, P/I/V, and Class cells emphasize in red to indicate that these cells require you to enter a value. Select the IND Class. Enter the numerical value of the inductor in Henries in the Value column. Enter the tolerance value of the inductor in the Tolerance column. Enter the rated current in mA in the P/I/V column.

### Ferrite Beads

When you select the Pin Type I, the Pin Mate, Value, Tolerance, P/I/V, and Class cells emphasize in red to indicate that these cells require you to enter a value. Select the BEAD Class. Enter the characteristic impedance Zo @ 100 MHz of the ferrite bead into the Value column, and the tolerance into the Tolerance column. Enter the rated current in mA in the P/I/V column.

## Power and Ground Pins

You can identify the power supplies and ground pins in the datasheet.

For input (PWR in type) and output (PWROUT pin type) power supplies, determine the values of Vmax and Vmin in the Recommended Operating Conditions provided by the device's datasheet. Sometimes the Features or Description section of the first page of the datasheet states the supply operation range.

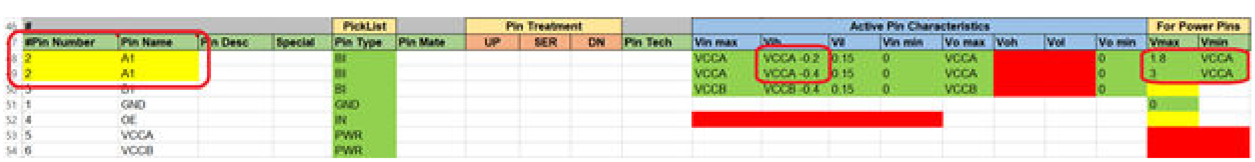
## Instantiations

An instantiation is a different threshold voltage for each different supply rail voltage.

In a datasheet, one instantiation of an A-port I/Os will have a range of 1.65 V to 1.95 V and a second instantiation of the A-port I/Os will have a range of 2.3 V to 3.6 V.

Figure 2-17 shows a model example of these datasheet values. The values for instantiations you enter in the Vmax cell should be the nominal voltage closest to the standard supply rail. For example, the instantiation values entered into the Vmax cell in this case is 1.8 V and 3 V, with the respective supply voltage rail VCCA in the Vmin cell. In the analysis, the HyperLynx Schematic Analysis tool looks at the instantiation value closest to the reference supply rail value.

**Figure 2-17: Model Example of Instantiations**



| # | | | | PickList | | Pin Treatment | | | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | UP | SER | DN | Pin Tech | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 2 | A1 | | | BI | | | | | | VCCA | VCCA -0.2 | 0.15 | 0 | VCCA | | | 0 | 1.8 | VCCA |
| 2 | A1 | | | BI | | | | | | VCCA | VCCA -0.4 | 0.15 | 0 | VCCA | | | 0 | 3 | VCCA |
| 3 | B1 | | | BI | | | | | | VCCB | VCCB -0.4 | 0.15 | 0 | VCCB | | | 0 | | |
| 1 | GND | | | GND | | | | | | | | | | | | | | 0 | |
| 4 | OE | | | IN | | | | | | | | | | | | | | | |
| 5 | VCCA | | | PWR | | | | | | | | | | | | | | | |
| 6 | VCCB | | | PWR | | | | | | | | | | | | | | | |

# Intentional Errors

The analysis can output a "CHECK" error in the results section by specifying "FLAG:" in the Comment column of a pin in a model.

This is meant for those occasions when there is a pin requirement that needs to be checked but it is not supported in the model. Since PWR pins do not support modeling of pin treatment requirements, a FLAG: comment can be added stating "FLAG: DVDD should be connected to VCC1 using a 1-2.2 Ohm resistor" when the part requires DVDD to be connected to VCC1 through filter resistor.

For example, Figure 2-21 shows the used of the "FLAG:" comment and the corresponding CHECK error that is generated during the analysis.

**Figure 2-18: Example of Intentional Check Error**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | Vmax | Vmin | Pin Comment |
|---|---|---|---|---|---|---|---|---|
| 4 | DVDD | | 1 | PWR | | 1.95 | 1.71 | FLAG: Connect a 1-2.2Ohm resistor from DVDD to VCC1. |
| 21 | AVDD | | 2 | PWR | | 1.95 | 1.71 | FLAG: Connect a 1-2.2Ohm resistor from AVDD to VCC1. |

| Severity | ID | Error | Description |
|---|---|---|---|
| BAD DATA | 2856 | CHECK | Pin U18400-21 (AVDD): Connect a 1-2.2Ohm resistor from AVDD to VCC1. |
| BAD DATA | 2859 | CHECK | Pin U18400-4 (DVDD): Connect a 1-2.2Ohm resistor from DVDD to VCC1. |

# 3. Advanced Modeling Examples

This chapter provides advanced modeling guidelines, tips, and examples to help you model various components and functions in a way that generates the best results from the schematic analysis, and reduces the quantity of false errors.

**Note**

The procedures in this chapter assume that you are already familiar with the modeling concepts described in the previous chapters ***Creating HyperLynx Schematic Analysis Models***and **Model Template Contents**.



## Modeling DDR4 DQ Thresholds

DDR4 memory uses the POD12 (Pseudo-Open Drain) I/O standard for the data bus. The address and command pins use the SSTL-12 I/O standard.

On DDR4 DRAM devices, the V data reference supply was removed from the package and is now internally generated by the DRAM. This means that the VREFDQ reference voltage can be set to any value over a wide range. This also means that the DRAM controller must set the DRAM's VREFDQ voltage to the proper value, matching its own reference voltage. This is accomplished through VREFDQ calibration.

The VREFDQ voltage can be set to one of two ranges using the DRAM's MR6 register. Range 1 is between 60% and 92.5% of VDDQ, and range 2 is between 40% and 77.5% of VDDQ.

The goal of the calibration is to find the best VREFDQ setting that sets the internal VREFDQ level to be the same as the DRAM's DQ midpoint. This is accomplished by determining which setting provides the largest optimal level for DQ and the lowest optimal level for DQ, and then using the level halfway in-between.

Having an internal VREFDQ reference level, of an unknown value, presents a modeling challenge for HyperLynx Schematic Analysis. The HyperLynx Schematic Analysis team has found several references that place the nominal VREFDQ level, or VREF level, for the POD12 I/O standard at 70% of VDDQ. This represents the default uncalibrated VREFDQ level.

- The first reference in **Table 3-1** represents typical JEDEC POD12 specifications.
- The second reference in **Table 3-2** represents typical Intel® Arria® 10 POD12 specifications.

**Table 3-1: JEDEC JESD8-24 POD12 DC Operating Conditions**

| Parameters | Symbol | POD12 | | | Units |
|---|---|---|---|---|---|
| | | Min | Typ | Max | |
| Data Supply Voltage | VDDQ | 1.16 | 1.2 | 1.26 | V |
| Reference Voltage | VREFDQ | 0.69 * VDDQ | 0.7 * VDDQ | 0.71 * VDDQ | V |
| DC Input Logic High Voltages | VIH(DC) | VREFDQ +0.08 | | VDDQ +0.15 | V |
| DC Input Logic Low Voltages | VIL(DC) | -0.15 | | VREFDQ -0.08 | V |

**Table 3-2: Single-Ended SSTL and POD12 Reference Voltage Specifications for Intel Arria 10 Devices**

| I/O Standards | VCCIO (V) | | | VREFDQ (V) | | |
|---|---|---|---|---|---|---|
| | Min | Typ | Max | Min | Typ | Max |
| SSTL-12 / SSTL-12 Class I, II | 1.14 | 1.2 | 1.26 | 0.49 x VCCIO | 0.5 x VCCIO | 0.51 x VCCIO |
| POD12 | 1.16 | 1.2 | 1.24 | 0.69 x VCCIO | 0.7 x VCCIO | 0.71 x VCCIO |

**Table 3-1** also defines the input logic high and input logic low levels relative to the VREFDQ level.

- Input logic high (VIH) minimum voltage is "VREFDQ + 0.08V".
- Input logic low (VIL) maximum voltage is "VREFDQ - 0.08V".

The driver output high and low logic levels are not as clearly defined. Typical JEDEC POD12 output levels are shown in **Table 3-3**.

**Table 3-3: Typical JEDEC Single-Ended AC & DC Output Levels**

| Parameters | Symbols | DDR4-1600-3200 | Units |
|---|---|---|---|
| DC output high measurement level (for IV curve linearity) | VOH(DC) | 1.1 x VDDQ | V |
| DC output mid measurement level (for IV curve linearity) | VOM(DC) | 0.8 x VDDQ | V |
| DC output low measurement level (for IV curve linearity) | VOL(DC) | 0.5 x VDDQ | V |
| AC output high measurement level (for output SR) | VOH(AC) | (0.7 + 0.15) x VDDQ | V |
| AC output low measurement level (for output SR) | VOL(AC) | (0.7 - 0.15) x VDDQ | V |

Since the DC specifications are for measuring the IV curve linearity of the ODT (On Die Termination) after proper ZQ calibration, we model the VOH and VOL AC specs defined for measuring output SR (Slew Rate) as follows:

- For VOH, (0.7 + 0.15) x VDDQ becomes “VDDQ *0.85”.
- For VOL, (0.7 - 0.15) x VDDQ becomes “VDDQ *0.55”.

Intel, for Arria 10 devices, specifies the VOH and VOL levels this way for POD12 drivers as shown in **Table 3-4**.

**Table 3-4: Single-Ended SSTL and POD12 I/O Voltage Specifications for Intel Arria 10 Devices**

| I/O Standards | VIL(DC) (V) | | VIH(DC) (V) | | VOL (V) | VOH (V) |
|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | Min | Max |
| HSTL-12 Class II | -0.15 | VREFDQ -0.08 | VREFDQ +0.08 | VCCIO +0.15 | 0.25 x VCCIO | 0.75 x VCCIO |
| POD12 | -0.15 | VREFDQ -0.08 | VREFDQ +0.08 | VCCIO +0.15 | (0.7 - 0.15) x VCCIO | (0.7 + 0.15) x VCCIO |

In order to be able to add the input specifications into the model, we need to define VREFDQ. However, because DDR4 memory does not have a VREFDQ pin, we instead need to define a virtual pin. A virtual pin is a pin without a pin number, but with both a pin name and a pin type of PWR.

**Figure 3-1** shows how all of these DQ specifications discussed so far are represented in a sample model.

**Figure 3-1: Sample Model For DDR4 DQ POD12 Pins**

| # | | | PickList | | | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | P | Pin Type | Pin Mate | | Pin Tech | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 1 | DQ0 | | DATA | 0_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 2 | DQ1 | | DATA | 1_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 3 | DQ2 | | DATA | 2_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 4 | DQ3 | | DATA | 3_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 5 | DQ4 | | DATA | 4_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 6 | DQ5 | | DATA | 5_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 7 | DQ6 | | DATA | 6_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 8 | DQ7 | | DATA | 7_8 | | POD12 | VDDQ | VREFDQ +0.08 | VREFDQ -0.08 | 0 | VDDQ | VDDQ *0.85 | VDDQ *0.55 | 0 | | |
| 20 | VDDQ | | PWR | | | | | | | | | | | | 1.26 | 1.14 |
| | VREFDQ | | PWR | | | | | | | | | | | | VDDQ *0.71 | VDDQ *0.69 |

Some vendors may use a different offset for their DDR4 DQ drivers. Xilinx® Zynq® UltraScale+ PS DDR4 uses 80% of the VCCO_PSDDR supply voltage, as shown in **Table 3-5**, instead of the 70% of VDDQ as shown above.

**Table 3-5: Xilinx Zynq UltraScale+ PS DDR DC Input and Output Levels Specifications**

| DDR Standard | VIL (V) | | VIH (V) | | VOL (V) | VOH (V) |
|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | Min | Max |
| DDR4 | 0.000 | VREFDQ-0.100 | VREFDQ +0.100 | VCCO_PSDDR | 0.8 x VCCO_PSDDR -0.150 | 0.8 x VCCO_PSDDR +0.150 |
| LPDDR4 | 0.000 | VREFDQ-0.100 | VREFDQ +0.100 | VCCO_PSDDR | 0.8 x VCCO_PSDDR -0.150 | 0.8 x VCCO_PSDDR +0.150 |
| DDR3 | -0.300 | VREFDQ-0.100 | VREFDQ +0.100 | VCCO_PSDDR | 0.8 x VCCO_PSDDR -0.175 | 0.8 x VCCO_PSDDR +0.175 |

For compatibility with existing models, these need to be modeled as 70% of VDDQ, or VCCO_PSDDR in this case.

- For VOH, “0.8 x VCCO_PSDDR + 0.150” becomes “VCCO_PSDDR *0.7+0.150”.
- For VOL, “0.8 x VCCO_PSDDR - 0.150” becomes “VCCO_PSDDR *0.7-0.150”.

In actual use, the receiver will go through VREFDQ calibration and will calibrate to 80% of VDDQ, or VCCO_PSDDR.

The DQ DC input thresholds are modeled as shown in **Figure 3-2**.

**Figure 3-2: Sample Model for Xilinx Zynq UltraScale+ DDR4 DC Input Thresholds**

| # | | PickList | | Active Pin Characteristics | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pin Type | Pin Mate | Vin max | Vih | Vil | Vin min | Vmax | Vmin |
| 1 | DQ0 | DATA | 0_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 2 | DQ1 | DATA | 1_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 3 | DQ2 | DATA | 2_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 4 | DQ3 | DATA | 3_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 5 | DQ4 | DATA | 4_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 6 | DQ5 | DATA | 5_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 7 | DQ6 | DATA | 6_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 8 | DQ7 | DATA | 7_8 | VCCO_PSDDR | VREFDQ +0.100 | VREFDQ -0.100 | 0 | | |
| 20 | VCCO_PSDDR | PWR | | | | | | 1.26 | 1.14 |
| | VREFDQ | PWR | | | | | | VCCO_PSDDR *0.71 | VCCO_PSDDR *0.69 |

The DC output thresholds are modeled, as shown in **Figure 3-3**, with the conversion to 70% of VDDQ.

**Figure 3-3: Sample Model for Xilinx Zynq UltraScale+ DDR4 DC Output Thresholds**

| # | | PickList | | Active Pin Characteristics | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pin Type | Pin Mate | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 1 | DQ0 | DATA | 0_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 2 | DQ1 | DATA | 1_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 3 | DQ2 | DATA | 2_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 4 | DQ3 | DATA | 3_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 5 | DQ4 | DATA | 4_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 6 | DQ5 | DATA | 5_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 7 | DQ6 | DATA | 6_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 8 | DQ7 | DATA | 7_8 | VCCO_PSDDR | VCCO_PSDDR *0.7+0.150 | VCCO_PSDDR *0.7-0.150 | 0 | | |
| 20 | VCCO_PSDDR | PWR | | | | | | 1.26 | 1.14 |
| | VREFDQ | PWR | | | | | | VCCO_PSDDR *0.71 | VCCO_PSDDR *0.69 |

# Modeling Differential Logic Levels

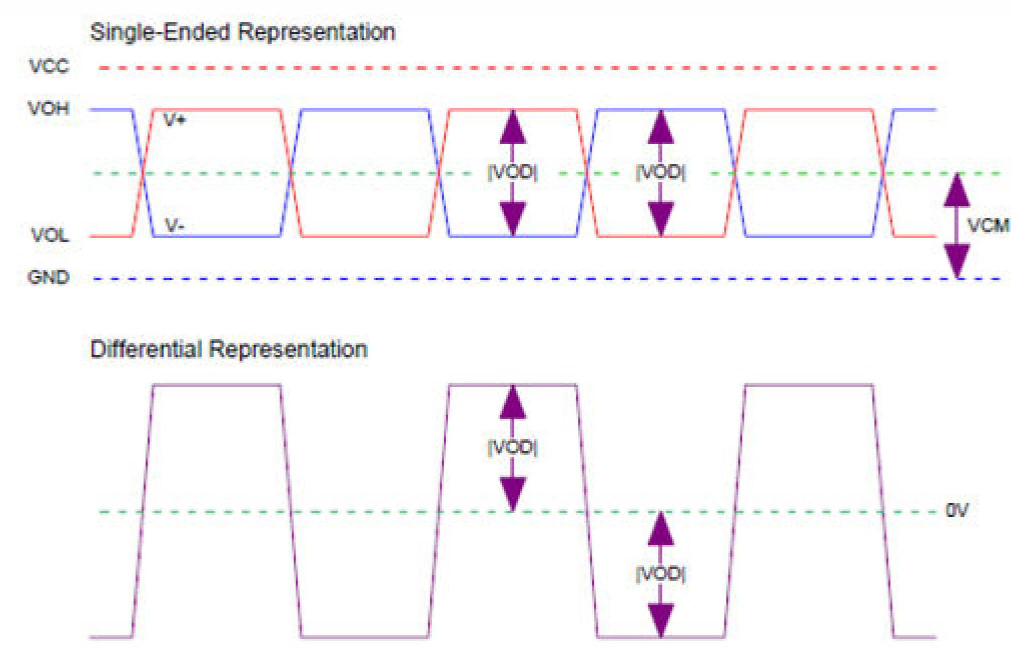
Differential signaling measures the difference between two complementary signals rather than the difference between a single signal and ground. For the receiving circuit to register a valid signal, the driver output differential voltage must be below the maximum differential input voltage (VID max), and above the minimum threshold (VID min).

The intention of this topic is not to teach differential signaling, but to show you how to model the differential logic levels for compatibility with existing models. For additional information on modeling differential logic levels, please refer to the *HyperLynx Schematic Analysis Model Library Guide and Reference*.

HyperLynx Schematic Analysis has adopted modeling the differential VID/VOD min and max levels as peak-to-peak (pk-pk) levels. The peak-to-peak levels are two times the absolute difference |VID| or |VOD| levels.

Consider the differential driver waveforms in **Figure 3-4**.

**Figure 3-4: Typical Differential Driver Waveforms**




The Single-Ended Representation shows the differential driver single-ended output levels centered on a common mode voltage VCM. The |VOD| level represents the absolute voltage difference between the V+ and V- waveforms and is considered the differential peak voltage.

In one phase, V+ swings above V- by |VOD|, and in the other phase V+ swings below V- by |VOD|. This is considered the peak differential level. If we normalize the V- waveform to 0V, this will result in a Differential Representation of the waveform as shown in **Figure 3-4**. This is also the way you would see the waveform on an oscilloscope when viewed using differential probes, and it is considered the peak-to-peak differential level.

The HyperLynx Schematic Analysis tool treats the differential receiver levels the same way as the differential driver levels. That is, for modeling purposes:

- VIN max is the maximum recommended DC input voltage that can be applied.
- VIH is the maximum peak-to-peak differential input voltage.
- VIL is the minimum peak-to-peak differential input voltage.
- VIN min is the minimum recommended DC input voltage that can be applied (typically 0V).
- VO max is the maximum DC output voltage that can be output from the pin.
- VOH is the maximum peak-to-peak differential output voltage.
- VOL is the minimum peak-to-peak differential output voltage.
- VO min is the lowest DC output voltage that can be driven by the pin (typically 0V).

For modeling and analysis purposes, HyperLynx Schematic Analysis ignores the VCM (common mode) voltage.

Most of the time, datasheets provide either peak differential levels or peak-to-peak levels. Sometimes they provide the single-ended levels that you will need to convert to differential levels.

- When provided with peak levels, double the values for VIH, VIL, VOH, and VOL.
- When provided with peak-to-peak levels, enter the values provided.
- When provided with single-ended (SE) levels, as shown in the Single-Ended Representation of **Figure 3-4**, convert them to differential levels as follows:
  - VIH = [(SE VIH max) - (SE Vil min)] *2
  - VIL = [(SE VIH min) - (SE VIL max)] *2
  - VOH = [(SE VOH max) - (SE VOL min) *2
  - VOL = [(SE VOH min) - (SE VOL max)] *2

## DDR4 Differential DQS Example

The JEDEC specification defines the DDR4 DQS Differential AC output levels as shown in **Table 3-6**. Because the specification does not provide any DC output levels, the AC output levels are modeled.

**Table 3-6: JEDEC Differential AC Output Level Specifications**

| Parameters | Symbols | DDR4-1600-3200 | Units |
|---|---|---|---|
| AC differential output high measurement level (for output SR) | VOHdiff(AC) | +0.3 x VDDQ | V |
| AC differential output low measurement level (for output SR) | VOLdiff(AC) | -0.3 x VDDQ | V |

**Table 3-6** defines VOH as “+0.3 x VDDQ” and VOL as "-0.3 x VDDQ”. The swing of “+/- 0.3 x VDDQ” , or “VDDQ *0.6”, represents the minimum differential output peak-to-peak swing, and is used as the Vol level.

The JEDEC JESD79-4B specification defines the DDR4 DQS differential input levels as shown in **Table 3-7**.

**Table 3-7: JEDEC Differential Input Levels for DQS**

| | | DDR4-1600-2133 | | DDR4-2400 | | |
|---|---|---|---|---|---|---|
| Parameters | Symbols | Min | Max | Min | Max | Units |
| Differential Input High | VIHDiff_DQS | 136 | | 130 | | mV |
| Differential Input Low | VILDiff_DQS | | -136 | | -130 | mV |

The minimum differential input voltage between DQS_t and DQS_c is “130 mV - (-130 mV)” or 260 mV for the 2400 MHz speed grade. Since these are specified as differential levels, 260 mV represents the minimum input peak-to-peak voltage required and is used as the Vil level.

The maximum input and output differential levels are not specified. We can assume that the peak input differential voltage cannot exceed the VDDQ supply. Similarly, the peak output voltage cannot be driven higher than the VDDQ driver power supply. Since these are peak differential levels, the peak-to-peak differential levels would be “VDDQ *2”.

**Figure 3-5** shows how a sample model represents all of the above differential DQS specifications for one of eight DQS pairs.

**Figure 3-5: Sample Model for Differential DQS Input and Output Levels**

| # | | | PickList | | | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | | Pin Tech | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 10 | DQS0_c | | DBIM | 11\|0_8 | | POD12 | VDDQ | VDDQ *2 | 0.26 | 0 | VDDQ | VDDQ *2 | VDDQ *0.6 | 0 | | |
| 11 | DQS0_t | | DBIP | 10\|0_8 | | POD12 | VDDQ | VDDQ *2 | 0.26 | 0 | VDDQ | VDDQ *2 | VDDQ *0.6 | 0 | | |
| 20 | VDDQ | | PWR | | | | | | | | | | | | 1.26 | 1.14 |

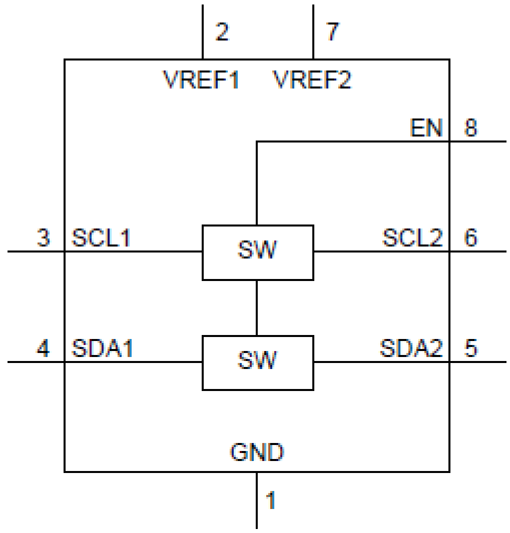
## Modeling Bidirectional Voltage Level Translators

Bidirectional voltage level translation devices enable translation without the use of a direction pin (DIR). These devices reduce the hardware effort because no logic is required to drive a DIR pin.

### Low ON-State Resistance Version

An example of a low ON-state resistance bidirectional voltage level translator is the PCA9306, which is available from various suppliers. It is mainly used in $I^2C$ and SMBus applications. **Figure 3-6** shows a functional block diagram of the PCA9306.

**Figure 3-6: PCA9306 Functional Block Diagram**

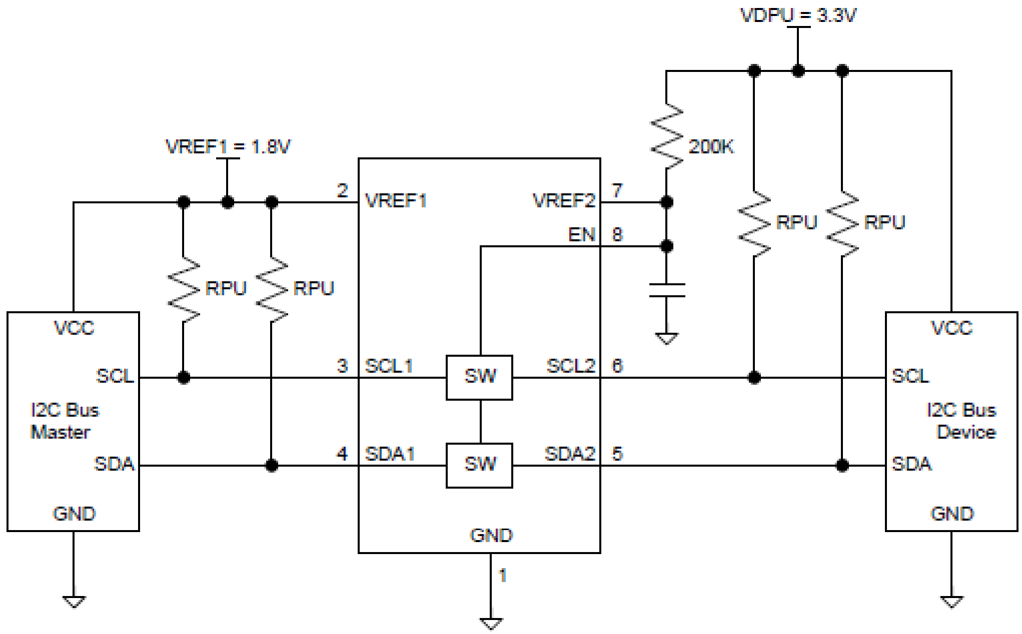



The low ON-state resistance of the switch (SW) minimizes propagation delay. When the enable input (EN) is high, the device connects the primary side (SCL1, SDA1) to the secondary side (SCL2, SDA2). When EN is low, a high impedance state exists between the two ports.

To support voltage level translation, VREF1 supports 1.2 V to “VREF2 - 0.6” V, and VREF2 must be between “VREF1 + 0.6” V to 5.5 V.

Figure 3-7 shows a typical application circuit of the PCA9306 device in which the switch is always enabled. For other applications, such as driving the EN input with a processor GPIO pin, please refer to the datasheet.

**Figure 3-7: PCA9306 Typical Application Circuit (Switch Always Enabled)**




The primary feature of the PCA9306 device is translating voltage from an I²C bus referenced to VREF1, up to an I²C bus referenced to VDPU, to which VREF2 is connected through a 200-kΩ pull-up resistor. To achieve translation on a standard, open-drain I²C bus, connect pull-up resistors from SCL1 and SDA1 to VREF1, and connect pull-up resistors from SCL2 and SDA2 to VDPU.

The PCA9306 device is translating voltage levels between the primary side (SCL1, SDA1) and the secondary side (SCL2, SDA2), which have different voltage references. Because the SW function is not linear over both reference voltage ranges, HyperLynx Schematic Analysis cannot use the GATE pin type, which is a linear function in which the input voltage always equals the output voltage, for translation function.

Instead, HyperLynx Schematic Analysis uses an AN (Analog Bidirectional) pin type with a pin mate assignment for the voltage level translation function. The pin mate allows the tool to test across the SW function between master and slave devices as it does for the GATE pin type. Since the AN pin type does not require a pin mate, the pin mate needs to be preceded by an asterisk ("*").

The following list summarizes how the tool models a Low ON-State Resistance auto-direction-sensing bidirectional voltage level translator device:

- The primary side of the device is referenced to VREF1.
- The secondary side of the device is referenced to VREF2.
- VREF1 supports 1.2 V to "VREF2 - 0.6V.
- VREF2 supports "VREF1 + 0.6" V to 5.5 V.

Figure 3-8 shows how a sample model represents these specifications:

**Figure 3-8: Sample Model for PCA9306 Voltage Level Translator**

| # | | | PickList | | Pin Tr | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | UP | | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 1 | GND | | GND | | | | | | | | | | | | 0 | |
| 2 | VREF1 | | PWR | | | | | | | | | | | | VREF2 -0.6 | 1.2 |
| 3 | SCL1 | | AN | *6 | | | VREF1 | | | 0 | VREF1 | | | 0 | | |
| 4 | SDA1 | | AN | *5 | | | VREF1 | | | 0 | VREF1 | | | 0 | | |
| 5 | SDA2 | | AN | *4 | | | VREF2 | | | 0 | VREF2 | | | 0 | | |
| 6 | SCL2 | | AN | *3 | | | VREF2 | | | 0 | VREF2 | | | 0 | | |
| 7 | VREF2 | | PWR | | | | | | | | | | | | 5.5 | VREF1 +0.6 |
| 8 | EN | | AIN | | R200K | | VREF2 | | | 0 | | | | | | |

## Edge-Rate Acceleration Version

Another example of an auto-direction-sensing bidirectional voltage level translator features a pass-gate architecture with edge-rate accelerators (one-shot) to improve the overall data rate. Examples are the Texas Instruments TCA9406 and NXP NTS0102. **Figure 3-9** shows the architecture of one I/O channel.

**Figure 3-9: Voltage Level Translator I/O Channel Architecture**

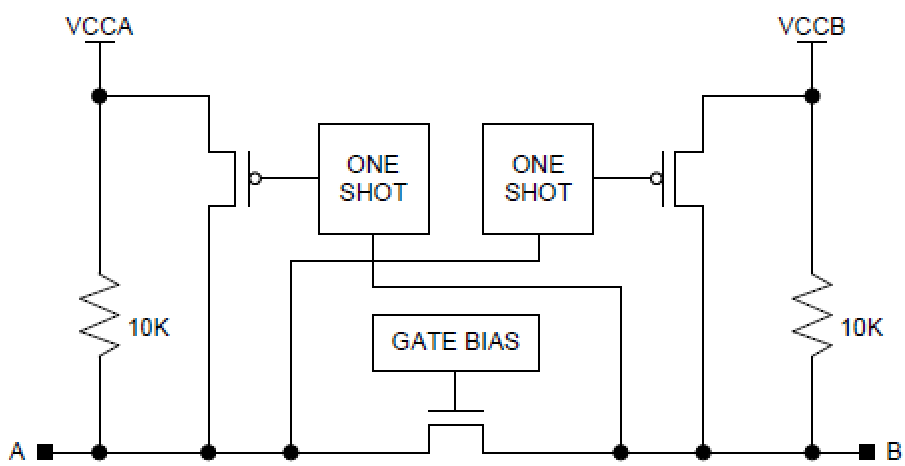



**Figure 3-9** shows dual supply voltage level translators featuring internal 10-kΩ pullup resistors. Additional external pullup resistors can be added to speed up rising edges. Edge-rate acceleration translators are targeted for use with open-drain applications, such as $I^2C$ and SMBus, but they can also be used in push-pull applications.

As shown in **Table 3-8**, the edge-rate accelerator version of the bidirectional level translator has input voltage threshold specifications that need to be met for proper operation.

**Table 3-8: Voltage Level Translator Input Threshold Specifications**

| Symbols | Ports | VCCA | VCCB | Min | Max | Units |
|---|---|---|---|---|---|---|
| VIH | A | 1.65 V to 1.95 V | 2.3 V to 5.5 V | VCCA -0.2 | VCCA | V |
| | | 2.3 V to 3.6 V | | VCCA -0.4 | VCCA | |

**Table 3-8: Voltage Level Translator Input Threshold Specifications (continued)**

| Symbols | Ports | VCCA | VCCB | Min | Max | Units |
|---|---|---|---|---|---|---|
| | B | 1.65 V to 1.95 V | 2.3 V to 5.5 V | VCCB -0.4 | VCCB | |
| | OE | | | VCCA x 0.65 | 5.5 | |
| VIL | A | 1.65 V to 1.95 V | 2.3 V to 5.5 V | 0 | 0.15 | V |
| | B | | | 0 | 0.15 | |
| | OE | | | 0 | VCCA x 0.35 | |

The power supply voltage range is 1.65 V to 3.6 V for port A (VCCA) and 2.3 V to 5.5 V for port B (VCCB) with the requirement that VCCA ≤ VCCB. The modeling of both requirements, at the same time, is not supported. We recommend modeling the voltage range with a comment that VCCA ≤ VCCB.

The edge-rate accelerator version of the bidirectional level translator also has output voltage threshold specifications as shown in **Table 3-9**.

**Table 3-9: Voltage Level Translator Output Threshold Specifications**

| Symbols | Ports | VCCA | VCCB | Min | Max | Units |
|---|---|---|---|---|---|---|
| VOH | A | 1.65 V to 3.6 V | 2.3 V to 5.5 V | VCCA x 0.67 | 0.4 | V |
| | B | | | VCCB x 0.67 | 0.4 | |
| VOL | A | 1.65 V to 3.6 V | 2.3 V to 5.5 V | | 0.4 | V |
| | B | | | | 0.4 | |

Due to all of the input and output thresholds that need to be modeled for this type of voltage level translator device, HyperLynx Schematic Analysis uses the OD (Open-Drain Bidirectional) pin type for this type of voltage.

The following list summarizes how the tool models bidirectional voltage level translator devices with edge-rate acceleration circuits:

- Port A is referenced to VCCA and port B is referenced to VCCB.
- The output enable pin OE is referenced to VCCA and is active high.
- The VCCA range is 1.65 V to 3.6 V, and the VCCB range is 2.3 V to 5.5.
- Requirement that VCCA ≤ VCCB.
- 10 kΩ pull-ups on port A and B pins.
- Input thresholds are as shown in **Table 3-8**. Notice the voltage instantiation on port A pins using standard voltage levels.

- Output thresholds are as shown in **Table 3-9**.
- Use the OD pin type to model input and output thresholds.

**Figure 3-10** shows how a sample model represents these specifications:

**Figure 3-10: Sample Model for Edge-Rate Acceleration Voltage Level Translator**

| # | | | PickList | | Pin | | Active Pin Characteristics | | | | | | | | For Power Pins | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | UP | | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin | Pin Comment |
| 2 | GND | | GND | | | | | | | | | | | | 0 | | |
| 6 | OE | | IN | | A | | 5.5 | VCCA *0.65 | VCCA *0.35 | 0 | | | | | | | |
| 5 | SCL_A | | OD | | 10K | | VCCA | VCCA -0.2 | 0.15 | 0 | VCCA | VCCA *0.67 | 0.4 | 0 | 1.8 | VCCA | |
| 4 | SDA_A | | OD | | 10K | | VCCA | VCCA -0.2 | 0.15 | 0 | VCCA | VCCA *0.67 | 0.4 | 0 | 1.8 | VCCA | |
| 5 | SCL_A | | OD | | 10K | | VCCA | VCCA -0.4 | 0.15 | 0 | VCCA | VCCA *0.67 | 0.4 | 0 | 2.5 | VCCA | |
| 4 | SDA_A | | OD | | 10K | | VCCA | VCCA -0.4 | 0.15 | 0 | VCCA | VCCA *0.67 | 0.4 | 0 | 2.5 | VCCA | |
| 8 | SCL_B | | OD | | 10K | | VCCB | VCCB -0.4 | 0.15 | 0 | VCCB | VCCB *0.67 | 0.4 | 0 | | | |
| 1 | SDA_B | | OD | | 10K | | VCCB | VCCB -0.4 | 0.15 | 0 | VCCB | VCCB *0.67 | 0.4 | 0 | | | |
| 3 | VCCA | | PWR | | | | | | | | | | | | 3.6 | 1.65 | VCCA ≤ VCCB |
| 7 | VCCB | | PWR | | | | | | | | | | | | 5.5 | 2.3 | |

## Modeling LDO Linear Voltage Regulators

Low-dropout (LDO) linear regulators function only as step-down regulators, with the limitation that the output voltage cannot be higher than the input voltage minus the maximum dropout voltage across the PMOS FET pass element.

**Figure 3-11** shows a simplified functional block diagram of an adjustable output LDO linear voltage regulator with an Under Voltage Lockout feature. Many adjustable output LDO linear voltage regulators come with additional features such as current limiting and thermal shutdown. The output voltage is set using a resistor divider from the FB pin to the OUT pin, and the FB pin and GND.

**Figure 3-11: Simplified Adjustable Output LDO Linear Voltage Regulator Block Diagram**

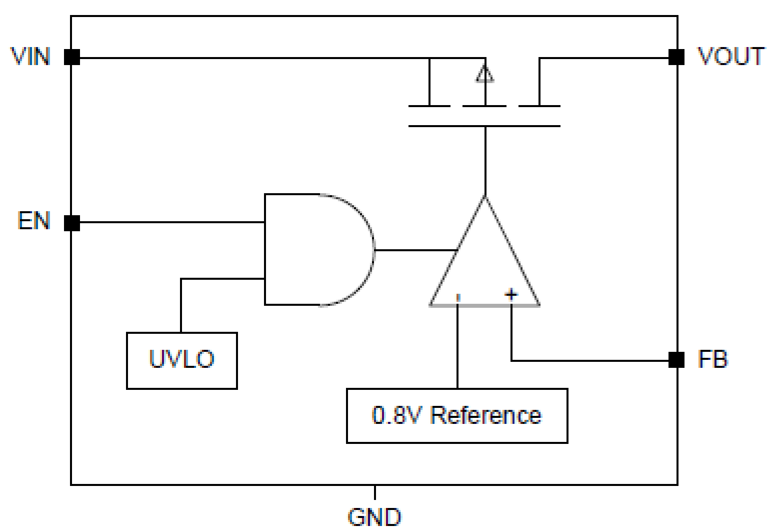



**Table 3-10** shows the typical specifications for an LDO linear voltage regulator.

**Table 3-10: LDO Linear Voltage Regulator Electrical Specifications**

| Symbol | Parameters | Test Conditions | Min | Typ | Max | Units |
|---|---|---|---|---|---|---|
| VIN | Input Voltage Range | | 2.5 | | 6.5 | V |
| VOUT | Output Voltage Range | | 0.9 | | 5 | V |
| VFB | Feedback Pin Voltage | IOUT = 5 mA | -2% | 0.8 | +2% | V |
| VDO | Dropout Voltage | IOUT = 150 mA | | 170 | 300 | mV |
| VIH | Enable High (Enabled) | | 1.25 | | 6.5 | V |
| VIL | Enabled Low (Disabled) | | 0 | | 0.4 | V |

The specifications of importance for modeling in **Table 3-10** are:

- The VIN input voltage range is from 2.5V to 6.5V.
- The maximum dropout voltage is 0.3V.
- The VOUT voltage range is 0.9V to 5V.
- The VFB voltage is 0.8V +/- 2% or 0.784V to 0.816V.
- The worst case dropout voltage is 300 mV.
- The EN pin input VIH and VIL thresholds are 1.25V and 0.4V respectively, with a max input voltage of 6.5V.

The enable pin, EN, when dot driven by other logic, is usually tied to the VIN pin on voltage regulators. If the Vin max level is modeled as VIN, instead of 6.5V, it will provide additional assurance that it is not mistakenly driven from unintended logic.

The max and min input and output voltages are clearly defined in the datasheet. There is also the additional requirement for linear voltage regulators that the input to output voltage meet the dropout voltage requirement. Specifying the full operating range of the input and output voltages, along with the dropout voltage, is not currently supported. We recommend the modeling of the dropout voltage for additional test coverage.

The dropout voltage requirement can be implemented in one of two ways:

- Included as part of the VIN pin Vmin specification as shown in **Figure 3-12**.

**Figure 3-12: Sample Model of LDO Linear Voltage Regulator With Dropout Voltage Applied to VIN Pin**

| # | | | PickList | | Pin Tr | | Active Pin Characteristics | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | UP | | Vin max | Vih | Vil | Vin min | Von | Vmax | Vmin |
| 1 | VIN | | PWR | | | | | | | | | 6.5 | VOUT +0.3 |
| 2 | GND | | GND | | | | | | | | | 0 | |
| 3 | EN | | IN | | A | | VIN | 1.25 | 0.4 | 0 | | | |
| 4 | FB | | FB | | | | | | | | | 0.816 | 0.784 |
| 5 | VOUT | | PWROUT | | | | | | | | | 5 | 0.9 |

- • Included as part of the VOUT pin Vmax specification as shown in **Figure 3-13**.

**Figure 3-13: Sample Model of LDO Linear Voltage Regulator With Dropout Voltage Applied to VOUT Pin**

| # | | | PickList | | Pin T | | Active Pin Characteristics | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | UP | | Vin max | Vih | Vil | Vin min | Vo in | Vmax | Vmin |
| 1 | VIN | | PWR | | | | | | | | | 6.5 | 2.5 |
| 2 | GND | | GND | | | | | | | | | 0 | |
| 3 | EN | | IN | | A | | 6.5 | 1.25 | 0.4 | 0 | | | |
| 4 | FB | | FB | | | | | | | | | 0.816 | 0.784 |
| 5 | VOUT | | PWROUT | | | | | | | | | VIN -0.3 | 0.9 |

## Modeling Load Switches

Load switches are electronic switches used to turn power rails on and off. When the device is enabled via the ON pin, the pass FET turns on, allowing current to flow from the input pin to the output pin, and power is passed to the downstream power rails.

**Figure 3-14** shows a diagram of a generic load switch.

**Figure 3-14: General Load Switch Circuit Diagram**

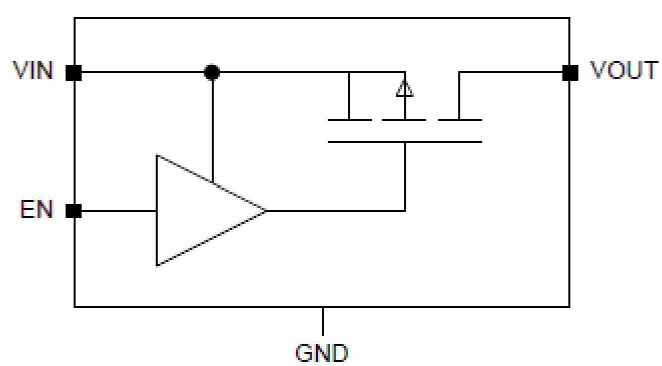



**Table 3-11** shows the relevant specifications for modeling a generic load switch.

**Table 3-11: Generic Load Switch Typical Electrical Specification**

| Symbols | Parameters | Min | Typ | Max | Units |
|---|---|---|---|---|---|
| VIN | Input Voltage Range | 2.5 | | 5.5 | V |
| VOUT | Output Voltage Range | 0 | | 5.5 | V |
| VIH | Enable High (Enabled) | 1.2 | | 5.5 | V |
| VIL | Enabled Low (Disabled) | 0 | | 0.4 | V |

From **Table 3-11**, the following information is useful for modeling purposes:

- The VIN input voltage range is 2.5V to 5.5V.
- The VOUT output voltage range is specified as 0 to 5.5V, but it actually passes through the input voltage.
- The ON pin input thresholds are as shown in **Table 3-11**.

Since the VOUT pin tracks the VIN power pin, it is best modeled as a power output (PWROUT) pin type. This helps to define the net connected to the VOUT pin as a voltage rail. Modeling the power output voltages on the VOUT pin, equal to the voltage on the VIN pin, helps the tool auto-detect the output voltages after defining the voltages on the VIN pin voltage net.

**Figure 3-15** shows how a sample model represents these specifications.

**Figure 3-15: Sample Model for Generic Load Switch**

| # | | | PickList | | Pin Tr | | Active Pin Charac | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | UP | | Vin max | Vih | Vil | Vin min | Vo n | Vmax | Vmin |
| 1 | VIN | | PWR | | | | | | | | | 5.5 | 2.5 |
| 2 | OUT | | PWROUT | | | | | | | | | VIN | VIN |
| 3 | ON | | IN | | A | | 5.5 | 1.2 | 0.4 | 0 | | | |
| 4 | GND | | GND | | | | | | | | | 0 | |

Many load switches have additional features, such as current limiting and BIAS voltages, that allow them to switch lower power supply voltages. You should model these as per the datasheet.

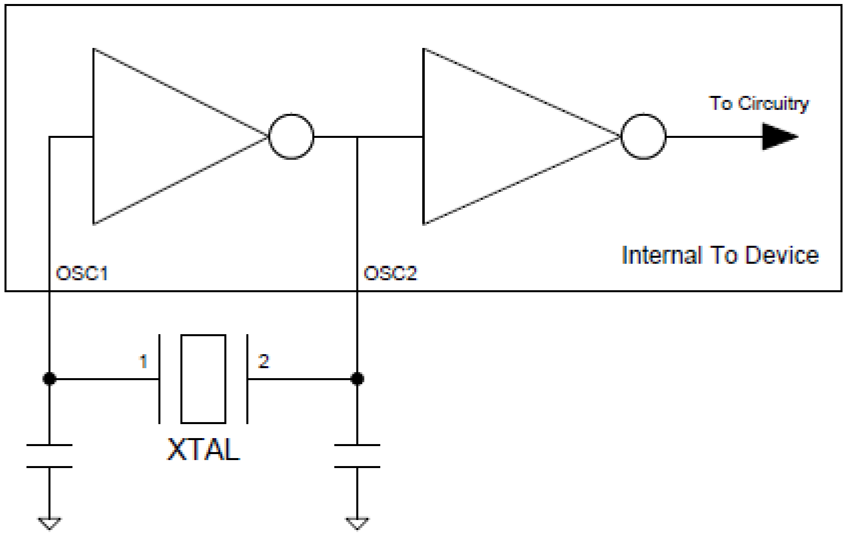
# Modeling Quartz Crystals and Crystal Oscillator Circuits

A typical crystal oscillator circuit is shown in the following figure. In this case, a quartz crystal is attached to a device whose internal circuitry consists of an inverting amplifier (inverter) providing a very high voltage gain and a 180 degree phase shift.

**Figure 3-16** shows a typical quartz crystal oscillator circuit.

**Figure 3-16: Typical Quartz Crystal Oscillator Circuit**




Because quartz crystals do not specify any DC electrical specifications, the HyperLynx Schematic Analysis tool uses the XTAL pin type that only requires a pin mate to be specified, as shown in the sample model in **Figure 3-17**. Some crystals provide two ground pins; if a crystal does have two ground pins, they also need to be modeled.

**Figure 3-17: Sample Quartz Crystal Model**

| # | | | PickList | | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 1 | XTAL1 | | XTAL | 2 | | | | | | | | | | | |
| 2 | XTAL2 | | XTAL | 1 | | | | | | | | | | | |

HyperLynx Schematic Analysis checks that one of the quartz crystal's XTAL pin types is connected to an input pin and that its pin mate is connected to an output pin. These are either analog (AIN and AOUT), if you do not provide thresholds, or digital (IN and OUT), if you do provide thresholds. The device datasheet specifies which is the input or output pin. **Figure 3-18** shows how a sample model represents both the analog and digital versions.

**Figure 3-18: Sample Device Quartz Crystal Interface**

| # | | | PickList | | Active Pin Characteristics | | | | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pin | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
| 11 | XTAL1 | | AIN | | VCC | | | 0 | | | | | | |
| 12 | XTAL2 | | AOUT | | | | | | VCC | | | 0 | | |
| 21 | OSC1 | | IN | | VCC | 2 | 0.8 | 0 | | | | | | |
| 22 | OSC2 | | OUT | | | | | | VCC | 2.4 | 0.4 | 0 | | |
| 30 | VCC | | PWR | | | | | | | | | | 3.465 | 3.135 |

# Modeling Dual MOSFETs with Common Pins

Power stages for switching power supply designs come in packages with dual MOSFETs. The common switch node of the two MOSFETs is internally connected and is also available externally. Due to the high current that the power stage switches, the common switch node is typically implemented on a single large pad or across multiple pins.

The dual MOSFET in Figure 3-19 has the common switch node, S1/D2, available on a single large pad-style pin (pin 5). Since this pin represents the Source pin of Q1 and the Drain pin of Q2, this pin needs to be instantiated twice in the model: once as an FETS pin type for Q1, and again as a FETD pin type for Q2.

**Figure 3-19: Typical Dual N-Channel MOSFET Power Stage with Single Common Pin**

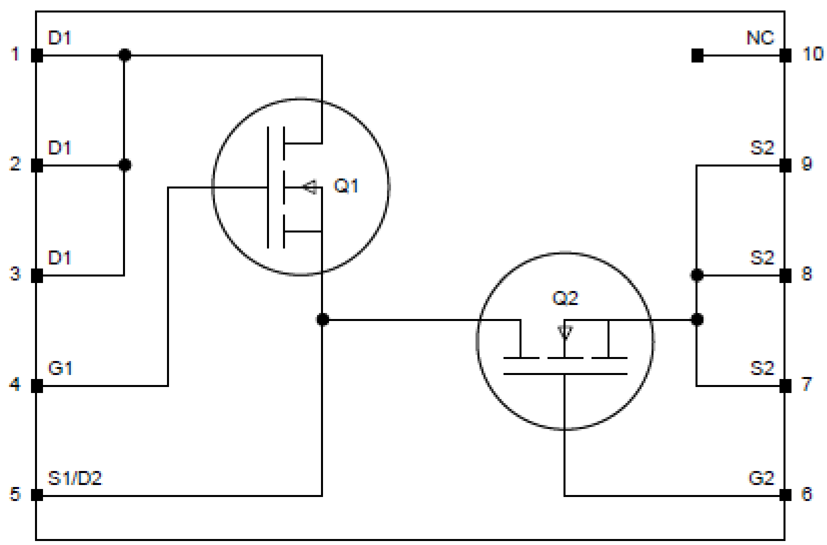



Figure 3-20 shows a sample model of the typical dual N-Channel MOSFET power stage with a single common pin.

**Figure 3-20: Sample Model for Dual N-Channel MOSFET Power Stage with Single Common Pin**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 2 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 3 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 4 | G1 | Q1 | | FETG | 5 | | 20 | 4.5 | 1 | 0 |
| 5 | S1 | Q1 | | FETS | | | | | | |
| 5 | D2 | Q2 | | FETD | 7 | | 30 | | | |
| 6 | G2 | Q2 | | FETG | 7 | | 20 | 4.5 | 1 | 0 |
| 7 | S2 | Q2 | | FETS | | | | | | |
| 8 | S2 | Q2 | | FETS | | | | | | |
| 9 | S2 | Q2 | | FETS | | | | | | |
| 10 | NC | | | NC | | | | | | |

Notice in Figure 3-20:

- The Pin Mate for all three Q1 D1 pins is the single Q1 S1 pin.
- The Pin Mate for the Q1 G1 pin is also the single Q1 S1 pin.
- The Q2 D2 and G2 pins can use any of the Q2 S2 pins as the Pin Mate. In this case, pin 7 was used.

The dual MOSFET in **Figure 3-21** has the common switch node, S1/D1, available on multiple pins. In this case, one of the S1/D2 pins can be applied to Q1, and the other can be applied to Q2.

**Figure 3-21: Typical Dual N-Channel MOSFET Power Stage with Multiple Common Pins**

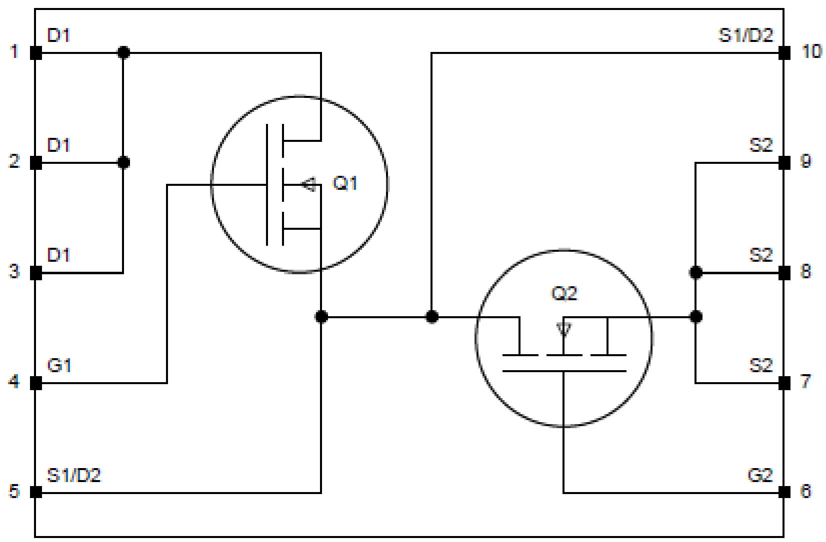



**Figure 3-22** shows a sample model of the typical dual N-Channel MOSFET power stage with two common pins. Pin 5 is used to represent the Source pin on Q1, and pin 10 is used to represent the Drain pin of Q2. Any additional pins for the common switch node S1/D2 can be applied to either Q1 or Q2.

**Figure 3-22: Sample Model for Dual N-Channel MOSFET Power Stage with Multiple Common Pins**

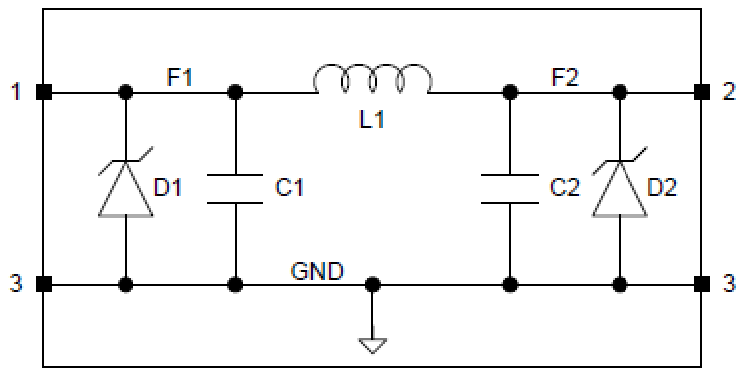
| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 2 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 3 | D1 | Q1 | | FETD | 5 | | 30 | | | |
| 4 | G1 | Q1 | | FETG | 5 | | 20 | 4.5 | 1 | 0 |
| 5 | S1 | Q1 | | FETS | | | | | | |
| 6 | G2 | Q2 | | FETG | 7 | | 20 | 4.5 | 1 | 0 |
| 7 | S2 | Q2 | | FETS | | | | | | |
| 8 | S2 | Q2 | | FETS | | | | | | |
| 9 | S2 | Q2 | | FETS | | | | | | |
| 10 | D2 | Q2 | | FETD | 7 | | 30 | | | |

# Modeling PI Filters

PI filters are primarily used in low pass filter applications. A PI filter is often also called an LC filter. The main components of a PI filter are the inductor and capacitor.

Sometimes, for off-board application use, protection diodes are also included as shown in **Figure 3-23**.

**Figure 3-23: Typical PI Filter With Protection Diodes**




Even though the common node of the PI filter is meant to be connected to GND, HyperLynx Schematic Analysis does not model this node or pin as GND. This allows us to fully model the protection diodes and capacitors, so that the tool can check if their rated voltages are being exceeded. **Figure 3-24** shows a typical model of the PI filter. Typical inductor, capacitor, and protection diode values were used. The protection diode is usually a unidirectional TVS.

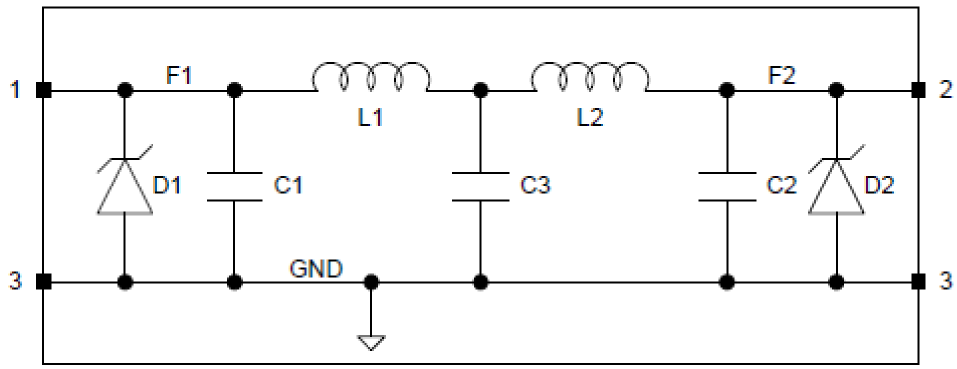
**Figure 3-24: Sample Model of Typical PI Filter With Protection Diodes**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | L1_1 | L1 | | I | 2 | | 15NH | 20 | 100 | IND |
| 2 | L1_2 | L1 | | I | 1 | | 15NH | 20 | 100 | IND |
| 1 | D1_1 | D1 | | DK | 3 | | 6 | 10 | 6.8 | TVS |
| 3 | D1_3 | D1 | | DA | 1 | | 6 | 10 | 6.8 | TVS |
| 2 | D2_2 | D2 | | DK | 3 | | 6 | 10 | 6.8 | TVS |
| 3 | D2_3 | D2 | | DA | 2 | | 6 | 10 | 6.8 | TVS |
| 1 | C1_1 | C1 | | C | 3 | | 10PF | 20 | 16 | CER |
| 3 | C1_3 | C1 | | C | 1 | | 10PF | 20 | 16 | CER |
| 2 | C2_2 | C2 | | C | 3 | | 10PF | 20 | 16 | CER |
| 3 | C2_3 | C2 | | C | 2 | | 10PF | 20 | 16 | CER |

Please refer to **Table 3-12** for modeling TVS specifications.

Some PI filters can have internal components and nodes that do not come out to external pins as shown in the back-to-back PI filter example of **Figure 3-25**. In this case, we cannot model capacitor C3. Since C3 has the same specifications as C1 and C2, it is effectively tested by the tests performed on C1 and C2. We also cannot model both inductors. In this type of case, you can add the inductance of L1 and L2, and model as the total inductance for pin 1 to pin 2.

**Figure 3-25: Typical Back-to-Back PI Filter With Protection Diodes**

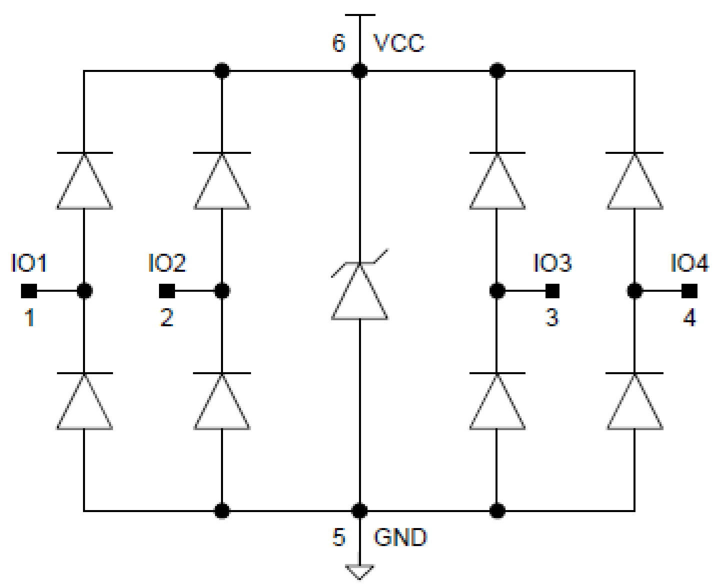



## Modeling ESD-Protection Devices

ESD-protection devices consist of an ESD-protection diode array to protect sensitive electronics attached to external communication lines, like USB or HDMI. Each IO channel consists of a pair of diodes that direct the ESD current pulses to VCC or GND.

Figure 3-26 shows a typical ESD-protection diode array based on the TPD6E001 device. This family also includes a TVS (Transient Voltage Suppressor) to clamp any voltage spikes that could be generated between VCC and GND. Since the diode type attached to the IO pins is not specified, it will be modeled as a Schottky diode.

**Figure 3-26: Typical ESD-Protection Diode Array**



Even though the common nodes of the ESD-protection diode array connect to VCC and GND, we do not model them as VCC and GND. We only model the individual diodes.

Table 3-12 shows the modeling requirements for the diode types used in the protection device shown in Figure 3-26.

**Table 3-12: Diode Modeling Requirements Versus Diode Type**

| Class | Type | Pin Mate | Value | Tol | P/I/V | Description |
|---|---|---|---|---|---|---|
| SCHOTTKY | DA, DK | Required | $VF_{(max)}$ | $IF_{(max)}$ | VR | Schottky |
| TVS | D, DA, DK | Required | VR[1] | $I_{pp}$@VC | VC | Transient Voltage Suppressor |
| Note: Current is modeled in mAmps.<br><br>For 2 pin devices, Pin 1 is the Cathode. | | | | | | |

The required Schottky and TVS specifications were extracted from the TPD6E001 datasheet and are summarized in Table 3-13.

**Table 3-13: ESD-Protection Diode Array Specifications**

| Symbol | Parameter | Test Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| VF | Diode forward voltage | IF = 10 mA | 0.65 | | 0.95 | V |
| VBR | Breakdown voltage | IBR = 10 mA | 11 | | | V |
| VCC | Supply voltage | | 0.9 | | 5.5 | V |
| ICC | Supply current | | | 1 | 100 | nA |
| VC | Channel clamp voltage | IF = 10 A | | | 25 | V |
| | | IF = 24 A | | | 60 | V |
| | | IF = 45 A | | | 100 | V |

Please note the following:

- Each IO pin has a diode connected to VCC and GND.

1 VR (Reverse Voltage) is sometimes referred to as Stand-off voltage or Working Reverse Voltage.

- There is also a TVS between VCC and GND.
- Since VCC has an operating range up to 5.5V, this is considered to be the working reverse voltage (VR) of the TVS.
- Since the TVS has three specifications for the clamp voltage (VC), we use the lowest value of 25V.
- The peak pulse current (IPP) at VC of 25V is 10A .

This model contains 9 diodes. It can be challenging to keep track of them all. To ensure that you do not overlook any of the diodes, you can use the Pin Desc column to number the diodes or use the pin numbers to help keep track of the diodes. You can also use the HyperLynx Schematic Analysis model importer to find missing pin mates should any instances of the diodes be missing from the model. The HyperLynx Schematic Analysis model importer generates an error if there are any missing pin mates and does not import the model.

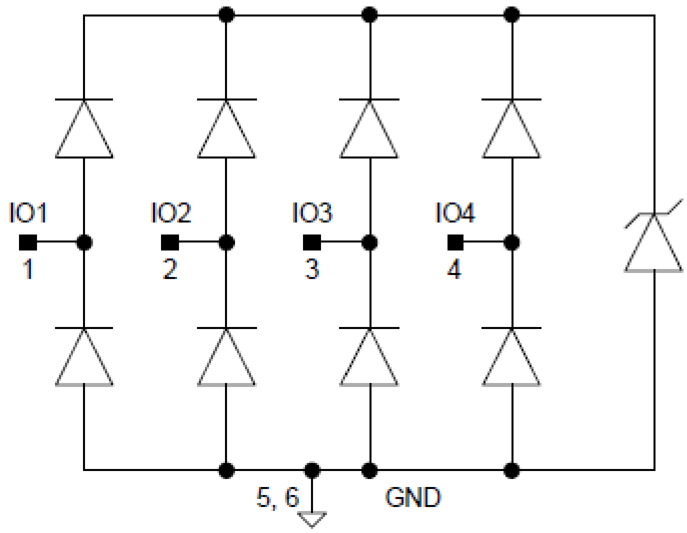
**Modeling ESD-Protection Devices**shows a sample model of the ESD-protection diode array represented in **Figure 3-26** using the diode specifications from **Table 3-13**.

**Figure 3-27: Sample Model of the ESD-Protection Diode Array**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | IO1 | D1-5 | | DK | 5 | | 0.95 | 10 | 11 | SCHOTTKY |
| 1 | IO1 | D1-6 | | DA | 6 | | 0.95 | 10 | 11 | SCHOTTKY |
| 2 | IO2 | D2-5 | | DK | 5 | | 0.95 | 10 | 11 | SCHOTTKY |
| 2 | IO2 | D2-6 | | DA | 6 | | 0.95 | 10 | 11 | SCHOTTKY |
| 3 | IO3 | D3-5 | | DK | 5 | | 0.95 | 10 | 11 | SCHOTTKY |
| 3 | IO3 | D3-6 | | DA | 6 | | 0.95 | 10 | 11 | SCHOTTKY |
| 4 | IO4 | D4-5 | | DK | 5 | | 0.95 | 10 | 11 | SCHOTTKY |
| 4 | IO4 | D4-6 | | DA | 6 | | 0.95 | 10 | 11 | SCHOTTKY |
| 5 | GND | D1-5 | | DA | 1 | | 0.95 | 10 | 11 | SCHOTTKY |
| 5 | GND | D2-5 | | DA | 2 | | 0.95 | 10 | 11 | SCHOTTKY |
| 5 | GND | D3-5 | | DA | 3 | | 0.95 | 10 | 11 | SCHOTTKY |
| 5 | GND | D4-5 | | DA | 4 | | 0.95 | 10 | 11 | SCHOTTKY |
| 6 | VCC | D1-6 | | DK | 1 | | 0.95 | 10 | 11 | SCHOTTKY |
| 6 | VCC | D2-6 | | DK | 2 | | 0.95 | 10 | 11 | SCHOTTKY |
| 6 | VCC | D3-6 | | DK | 3 | | 0.95 | 10 | 11 | SCHOTTKY |
| 6 | VCC | D4-6 | | DK | 4 | | 0.95 | 10 | 11 | SCHOTTKY |
| 5 | GND | TVS | | DA | 6 | | 5.5 | 10000 | 25 | TVS |
| 6 | VCC | TVS | | DK | 5 | | 5.5 | 10000 | 25 | TVS |

Another example of an ESD-protection device is the low capacitance TVS array, as shown in **Figure 3-28**, based on the RClamp052xPA device. Unlike the previous TPD6E001 ESD-protection device, the ESD-protection TVS array does not have a common node connected to VCC. The TVS array provides the over-voltage and under-voltage clamping functions relative to GND.

**Figure 3-28: Typical ESD Protection TVS Array**




The required diode/TVS specifications were extracted from the RClamp052xPA datasheet and are summarized in **Table 3-14**.

**Table 3-14: ESD-Protection TVS Array Specifications**

| Symbol | Parameter | Test Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| VRWM | Reverse Stand-Off Voltage | Any IO pin to GND | | | 5 | V |
| IR | Reverse Leakage Voltage | VRWM = 5V<br><br>Any IO pin to GND | | | 1 | µA |
| VC | Clamping Voltage | IPP = 1A<br><br>Any IO pin to GND | | | 15 | V |

Please note the following:

- The TVS + diode, between the IO and GND pins, provides the over-voltage protection (clamping) relative to GND.
- The single diode, between the IO and GND pins, provides the under-voltage protection, relative to GND.
- All IOs have a diode, and TVS with diode, to GND. We are not able to model both scenarios. For modeling purposes use TVS diode type.
- For the VR specification, use the VRWM (Reverse Stand-Off Voltage) specification.
- The clamp voltage, VC, is 25V.

- The peak pulse current (IPP) at VC of 15V is 1 A.

**Figure 3-29** shows a sample model of the ESD-protection TVS array represented in **Figure 3-28**, using the diode/TVS specifications from **Table 3-14**. All of the IO pins are referenced to both GND pins in the device. For modeling purposes, at least one of the IO pins needs to be modeled to each GND pin, and back to the IO pin, using the Pin Mate column. You can model the other IO pins to either GND pin.

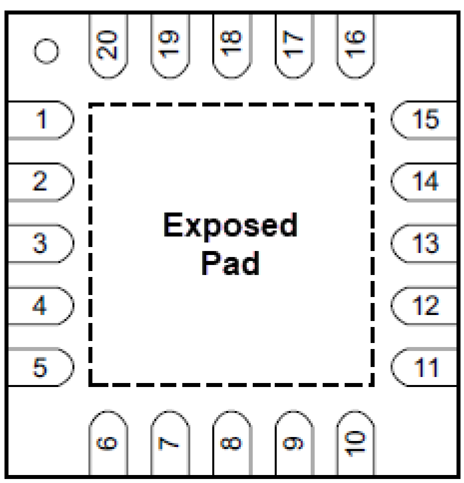
**Figure 3-29: Sample Model of the ESD-Protection TVS Array**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | IO1 | D1-5 | | DK | 5 | | 5 | 1000 | 15 | TVS |
| 2 | IO2 | D2-5 | | DK | 5 | | 5 | 1000 | 15 | TVS |
| 3 | IO3 | D3-6 | | DK | 6 | | 5 | 1000 | 15 | TVS |
| 4 | IO4 | D4-6 | | DK | 6 | | 5 | 1000 | 15 | TVS |
| 5 | GND | D1-5 | | DA | 1 | | 5 | 1000 | 15 | TVS |
| 5 | GND | D2-5 | | DA | 2 | | 5 | 1000 | 15 | TVS |
| 6 | GND | D3-6 | | DA | 3 | | 5 | 1000 | 15 | TVS |
| 6 | GND | D4-6 | | DA | 4 | | 5 | 1000 | 15 | TVS |

## Modeling Negative Voltage Exposed Pads

Many dual supply devices, with both positive and negative voltage rails, also have an exposed pad (or thermal pad) underneath the device as shown in the following figure.

**Figure 3-30: Sample Device Footprint with Exposed Pad**




When the exposed pad (EPAD) is internally connected to the negative power supply (VSS), the following modeling technique is recommended:

- Model the full operating range of the negative VSS supply with a PWRN pin type (-15V to 0V, 10% tolerance).

- Model the differential power supply range that is supported between the positive VDD supply and VSS. In this case, the minimum differential voltage between VDD and VSS is 9V and the maximum differential voltage is 33V.
- Apply the pin mate from VSS (PWRN pin type) to VDD.
- For the EPAD, use VSS as the **Vmax** and **Vmin** power supply requirement and use the same Pin Type that was used for the VSS pin.
- Optionally, add VSS in the Pin Desc column for the EPAD pin to remind you that the pin is internally connected to VSS or add a comment in the Comment column.

Figure 3-31 shows how this technique is implemented as part of a model.

**Figure 3-31: Sample Model of Device With Negative Voltage Exposed Pad**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | U nin | Vmax | Vmin |
|---|---|---|---|---|---|---|---|---|
| 10 | VSS | | | PWRN | 20 | | 0 | -16.5 |
| 20 | VDD | | | PWR | | | 33 | 9 |
| 21 | EPAD | VSS | | PWRN | 20 | | VSS | VSS |

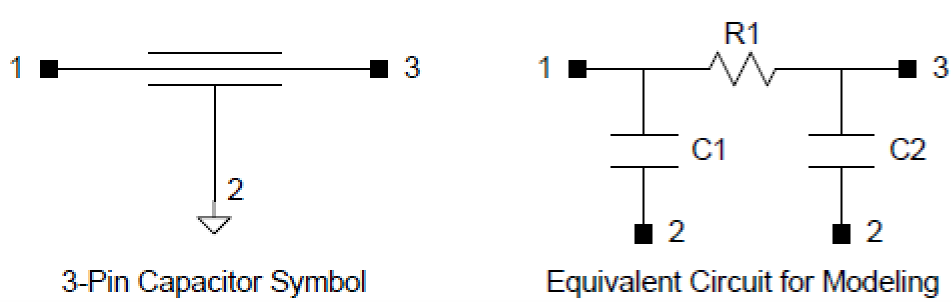
You could consider modeling the EPAD pin with the same voltage requirement as the VSS pin, but this is not recommended. While that would also work, HyperLynx Schematic Analysis would not catch cases where the EPAD was connected to a different negative supply voltage from VSS. For example, it would not catch the case where EPAD is connected to -5V and VSS is connected to -15V, as both voltages would meet the requirements of both pins.

## Modeling 3-Pin Capacitors

With the increasing speed in electronics, 3-pin feed-through filters, having lower ESL (equivalent series inductance), can be used for noise filtering or decoupling.

Figure 3-32 shows a typical symbol of a 3-pin capacitor and an equivalent circuit that can be used for modeling.

**Figure 3-32: 3-Pin Capacitor Symbol & Equivalent Circuit for Modeling**



The 3-pin capacitor has a feed-through component that can be modeled as a resistor and capacitive filter component usually connected to GND. Even though the common node, pin 2, is meant to be connected to GND we do not model this node as GND. This allows us to fully model the capacitors and apply the specifications of the capacitor from the datasheet. The resistor and capacitor values can be found in the datasheet. The resistor power can be calculated using the maximum current specification and the resistor value. If the resistor tolerance is not specified use ≤5%.

Figure 3-33 shows a sample model of a 3-pin capacitor using typical values.

**Figure 3-33: Sample Model of a Typical 3-Pin Capacitor**

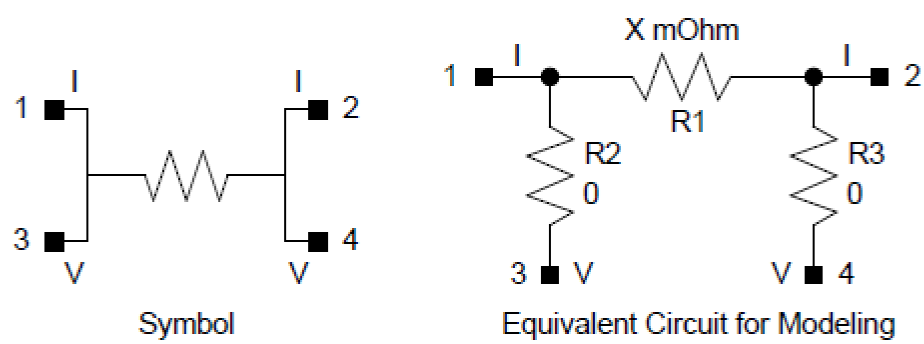
| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | R1_1 | R1 | | R | 3 | | 0.005 | 5 | 250 | |
| 3 | R1_3 | R1 | | R | 1 | | 0.005 | 5 | 250 | |
| 1 | C1_1 | C1 | | C | 2 | | 27U | 20 | 6.3 | CER |
| 2 | C1_2 | C1 | | C | 1 | | 27U | 20 | 6.3 | CER |
| 3 | C2_3 | C2 | | C | 2 | | 27U | 20 | 6.3 | CER |
| 2 | C2_2 | C2 | | C | 3 | | 27U | 20 | 6.3 | CER |

## Modeling Current Sense Resistors

Current sense resistors enable the measurement of current flow in a circuit by monitoring a voltage drop across a precisely calibrated resistance using four terminals. This configuration enables current to be applied through two opposite terminals and a sensing voltage to be measured across the other two terminals.

Figure 3-34 shows a typical symbol of a current sense resistor and an equivalent circuit that can be used for modeling.

**Figure 3-34: Current Sense Resistor Symbol & Equivalent Circuit for Modeling**




The resistance, tolerance power rating of the current sense path is specified in the datasheet. Since the voltage sense terminals are physically connected to the current terminals, they are modeled as 0 Ohm resistors. For the tolerance and power rating, the values of the current sense path resistor can be used.

Figure 3-35 shows a sample model of current sense resistor.

**Figure 3-35: Sample Model of a Current Sense Resistor**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | R1_1 | R1 | | R | 2 | | 0.003 | 1 | 1000 | |
| 2 | R1_2 | R1 | | R | 1 | | 0.003 | 1 | 1000 | |
| 1 | R2_1 | R2 | | R | 3 | | 0 | 1 | 1000 | |
| 3 | R2_3 | R2 | | R | 1 | | 0 | 1 | 1000 | |
| 2 | R3_2 | R3 | | R | 4 | | 0 | 1 | 1000 | |
| 4 | R3_4 | R3 | | R | 2 | | 0 | 1 | 1000 | |

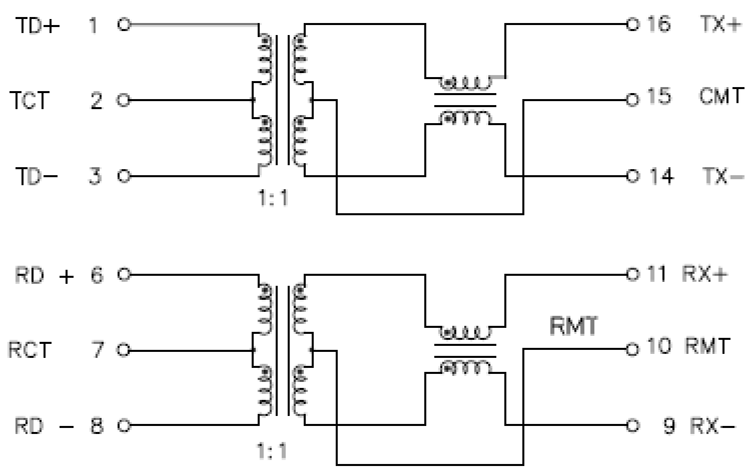
## Modeling Transformers

Transformers are modeled using the following pin types:

- CTAP: Transformer center tap pin.
- XFMR: Transformer multi-tap pin.

Figure 3-36 Shows a typical transformer with center tap pins.

**Figure 3-36: Typical Transformer Diagram**




Information for modeling purposes:

- Each center tap pin connects to 2 XFMR pins on the same side of the winding.
- The pin mate of each primary XFMR pin is the corresponding secondary XFMR pin, and vice versa.
- The turns ration is 1:1, primary to secondary, and is entered as 1 in the Vin max column for XFMR pin types.

The transformer is modeled as shown in Figure 3-37

**Figure 3-37: Sample Model of a Transformer**

| #Pin Number | Pin Name | Pin Desc | Special | Pin Type | Pin Mate | ch | Vin max | Vih | Vil | Vin min |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | RD_M | | | XFMR | 9 | | 1 | | | |
| 6 | RD_P | | | XFMR | 11 | | 1 | | | |
| 9 | RX_M | | | XFMR | 8 | | 1 | | | |
| 11 | RX_P | | | XFMR | 6 | | 1 | | | |
| 3 | TD_M | | | XFMR | 14 | | 1 | | | |
| 1 | TD_P | | | XFMR | 16 | | 1 | | | |
| 14 | TX_M | | | XFMR | 3 | | 1 | | | |
| 16 | TX_P | | | XFMR | 1 | | 1 | | | |
| 10 | RMT | | | CTAP | 9 | | | | | |
| 10 | RMT | | | CTAP | 11 | | | | | |
| 15 | CMT | | | CTAP | 14 | | | | | |
| 15 | CMT | | | CTAP | 16 | | | | | |
| 7 | RCT | | | CTAP | 6 | | | | | |
| 7 | RCT | | | CTAP | 8 | | | | | |
| 2 | TCT | | | CTAP | 1 | | | | | |
| 2 | TCT | | | CTAP | 3 | | | | | |

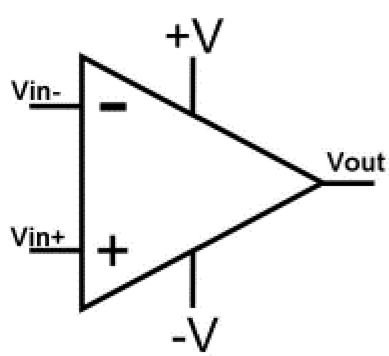
The primary to secondary turns ratio, N, is 1:1, and is entered in the Vin max column. The turn ratio back from the secondary to the primary is 1/N. In this case it is still 1.

If the primary to secondary turns ratio was 2, then the secondary to primary turn ratio would be ½, but would be modeled as 0.5.

# Modeling Dual Supply Operational Amplifiers

A typical dual voltage op-amp or operational amplifier is shown in Figure 3-38.

**Figure 3-38: Typical Dual Supply Op-Amp Block Diagram**




- Nominal power supply specs for a dual voltage op-amp are as shown in Table 3-15

**Table 3-15: Op-Amp Nominal Power Supply Specifications**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| V(supply) | Nominal Supply Voltage | +/-2.5 | | +/-15 | V |

Please note that these are nominal voltage levels and do not include tolerance. Most devices include a power supply tolerance, on the nominal voltage levels, as do the voltages defined in the Schematic Analysis voltages tab. In order to avoid potential false voltage errors, it is recommended to apply the typically used 5% tolerance to the nominal voltage levels. The full range of the allowed power supply range recommended for modeling is shown in **Table 3-16**.

**Table 3-16: Op-Amp Power Supply Specifications with Tolerance**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| V(supply) | Supply Voltage with 5% Tolerance | +/-2.375 | | +/-15.75 | V |

When the power supply specifications already include tolerance, the supply voltage range should be modeled as listed in the datasheet.

For dual supplies, we model the full voltage range for V-, and the supported difference voltage (V+ - V-) range for V+.

- The range on the V- supply is modeled as 0V to -15.75V.
    - This still allows V- to be connected to GND in case the op-amp is used in single supply mode.
    - It is modeled using the PWRN pin type with pin mate to the V+ pin.
- The range on the V+ supply is 4.75V to 31.5V and represents the "V+ - V-" difference voltage.
    - 4.75V is 2*2.35V and represents the minimum difference voltage supported.
    - 31.5 is 2*15.75V and represents the maximum difference voltage supported.

The inputs are modeled as AINM and AINP pin types with pin mates to the output (AOUT pin type). This allows the analysis to check that:

- The op-amp circuit does not have positive feedback.
- The gain is not excessive (greater than 1000).

The completed dual supply op-amp sample model, as per above specifications is shown in **Figure 3-39**

**Figure 3-39: Sample Dual Supply Op-Amp Model**

| #Pin Number | Pin Name | P | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | VOUT | | AOUT | | | | | | | V_P | | | V_M | | |
| 2 | VIN_M | | AINM | 1 | | V_P | | | V_M | | | | | | |
| 3 | VIN_P | | AINP | 1 | | V_P | | | V_M | | | | | | |
| 4 | V_M | | PWRN | 5 | | | | | | | | | | 0 | -15.75 |
| 5 | V_P | | PWR | | | | | | | | | | | 31.5 | 4.75 |

Sample model of a strictly single supply op-amp is shown in **Figure 3-40**.

**Figure 3-40: Sample Single Supply Op-Amp Model**

| #Pin Number | Pin Name | Pi | Pin Type | Pin Mate | | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min | Vmax | Vmin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | VOUT | | AOUT | | | | | | | V_P | | | 0 | | |
| 2 | VIN_M | | AINM | 1 | | V_P | | | 0 | | | | | | |
| 3 | VIN_P | | AINP | 1 | | V_P | | | 0 | | | | | | |
| 4 | V_M | | GND | | | | | | | | | | | 0 | |
| 5 | V_P | | PWR | | | | | | | | | | | 5.25 | 2.375 |

# Modeling Open-Drain/Collector Pins

There are two types of Open-Drain/Collector Pins:

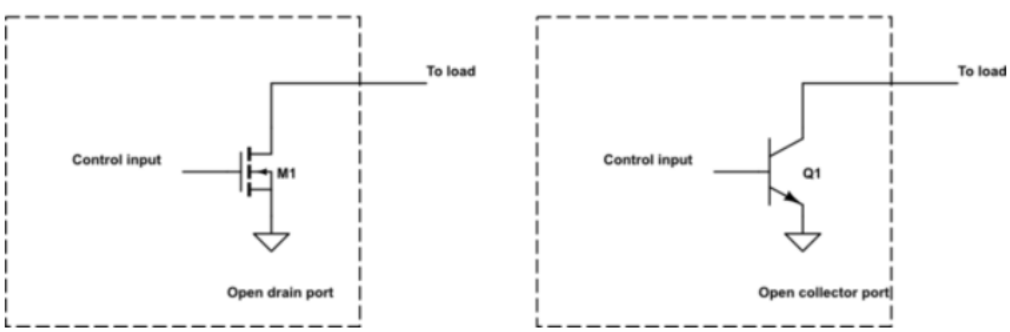
- Output only
- Bidirectional

## Modeling Open-Drain/Collector Output Pins

An open-drain or open-collector output pin is a transistor (either a MOSFET or BJT) with its drain or collector pin left floating (also known as open or high-impedance) and the source or emitter tied to ground as shown in the figure below.

A driver can turn the transistor on to pull the signal to ground or leave it open when the transistor is off. When the transistor is off, the signal can be pulled up or down by an exterrnal resistor to prevent an undefined, floating state.

**Figure 3-41: Open-Drain/Collector Output Pins**




Since the MOSFET drain and the BJT collector are floating their limitations are as follows:

- The limit on the MOSFET drain is equivalent to the MOSFET Vds spec.
- The limit on the BJT source is equivalent to the BJT Vce spec.

These can be obtained from the recommended operating conditions. Example limits are:

- MOSFET open-drain output pin max supported voltage of 5V.
- BJT open-collector output pin max supported voltage of 7V.

These represent the max power supply voltage supported as a pullup voltage:

- Vol is 0.4V

The open-drain and open-collector output pins, using ODOUT and OCOUT pin types, would be modeled as follows:

**Figure 3-42: Sample Model of Open-Drain/Collector Output Pins**

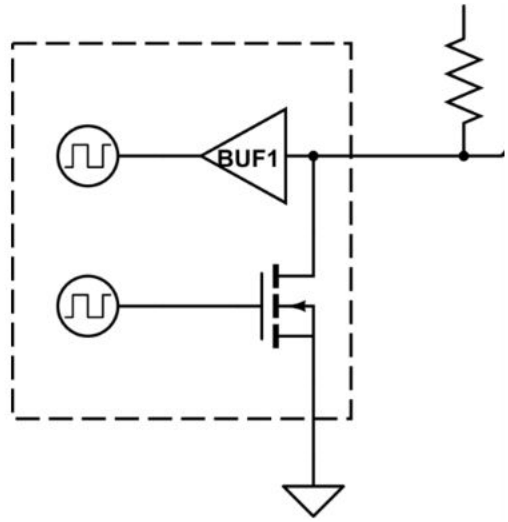
| # | | | PickList | | Active Pin Characteristics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pi | Pin Type | Pi | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min |
| 1 | ODOUT_PIN | | ODOUT | | | | | | 5 | 5 | 0.4 | 0 |
| 2 | OCOUT_PIN | | OCOUT | | | | | | 7 | 7 | 0.4 | 0 |

- Voh is modeled with the same level as the Vo max level.
    - If the datasheet specifies a Voh level, due to internal circuitry or pullup, then use that level for Voh.
- If the max supported voltage is not listed in recommended operating conditions, then use the value from the absolute maximum ratings.
    - If the maximum level is only specified as VCC or VDD, they use that as the Vo max spec.

## Modeling Bidirectional Open-Drain/Collector Pins

A bidirectional open-drain pin has a floating open-drain output component as well as an LVCMOS/TTL input component as shown in the figure below. It still requires and external pullup resistor.

**Figure 3-43: Bidirectional Open-Drain Pin**




Example input and output specifications for a bidirectional open-drain pin are shown in the table below.

**Table 3-17: Bidirectional Open-Drain Pin Specifications**

| Symbol | Min | Max | Unit |
|---|---|---|---|
| VIH | VDD x 0.7 | 5.5 | V |
| VIL | 0 | VDD x 0.3 | V |
| VOL | 0 | 0.4 | V |

The bidirectional open-drain pin, using OD pin type, would be modeled as follows:

**Figure 3-44: Sample Model of Bidirectional Open-Drain Pin with 5V Tolerant Input**

| # | | | PickList | | Active Pin Characteristics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | P | Pin Type | Pi | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min |
| 1 | OD_PIN | | OD | | 5.5 | VDD *0.7 | VDD *0.3 | 0 | 5.5 | 5.5 | 0.4 | 0 |

- The Vo max level is limited to the specification of the Vin Max level, which in this case is 5V tolerant.
- Voh is modeled with the same level as the Vo max level.
  - If the datasheet specifies a Voh level, due to internal circuitry or pullup, then use that level for Voh.

If the Vin max level was limited to VDD, and not 5V tolerant, then it would be modeled as:

**Figure 3-45: Sample Model of Bidirectional Open-Drain Pin with VDD Limited Input Voltage**

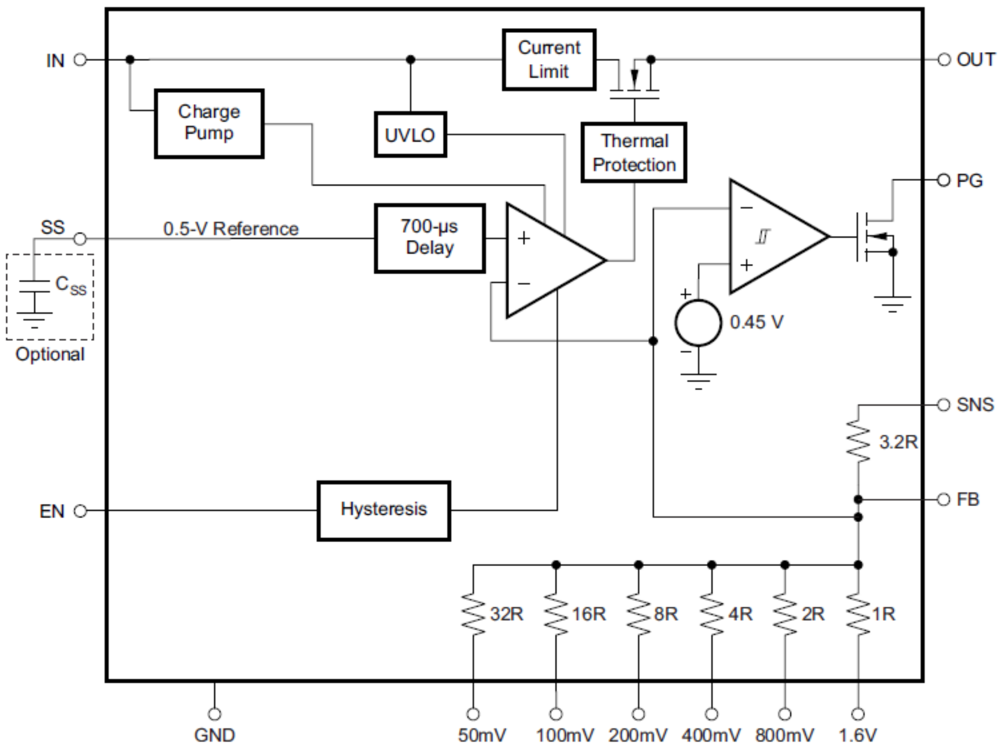
| # | | | PickList | | Active Pin Characteristics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | P | Pin Type | Pi | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min |
| 1 | OD_PIN | | OD | | VDD | VDD *0.7 | VDD *0.3 | 0 | VDD | VDD | 0.4 | 0 |

# Modeling Any Output Voltage Regulators

Any output voltage regulators have a user-configurable output voltage setting that does not require external FB resistors. The UP and DOWN FB resistors are built into the device as shown in Any Output Voltage Regulator Block Diagram.png.

The regulators support a wide range of output voltages, in 50mV or 100mV increments, depending on the device.

**Figure 3-46: Any Output Voltage Regulator Block Diagram**




Any output voltage regulators have the following requirements:

- For proper voltage setting, the FB 3.2R UP resistor on the SNS pin, needs to be connected to one of the OUT pins.
- The FB DOWN resistors, either need to be left floating, or tied to GND.

These requirements would be modeled as follows:

**Figure 3-47: Sample Model of Any Output Voltage Regulator**

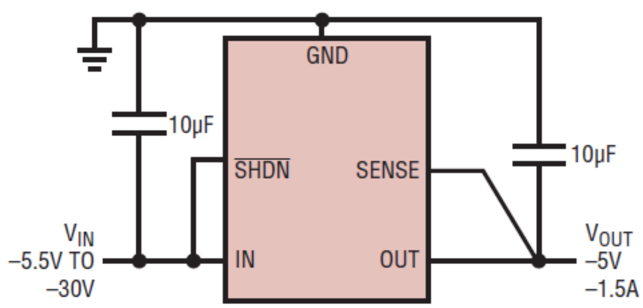
| # | | | PickList | | Pin Treatment | | | | Active Pin Characteristics | | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | P l | Pin Type | Pi | UP | SER | DN | Pi | Vin max | Vih | Vil | Vin min | Vo n | Vmax | Vmin |
| 2 | SNS | | AIN | | | @1 | | | 5 | | | 0.9 | | | |
| 5 | 50mV | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 6 | 100mV | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 7 | 200mV | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 9 | 400mV | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 10 | 800mV | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 11 | 1.6V | | AIN | | | N | | | 0.1 | | | 0 | | | |
| 1 | OUT | | PWROUT | | | | | | | | | | | IN -0.35 | 0.9 |
| 19 | OUT | | PWROUT | | | | | | | | | | | IN -0.35 | 0.9 |
| 20 | OUT | | PWROUT | | | | | | | | | | | IN -0.35 | 0.9 |

- The SNS pin is modeled with the supported output voltage range of the regulator.
- The Vin max limit on the DOWN resistor pins is modeled as 0.1V to test that when it is connected, it is connected to GND, and N in the SER column as it can optionally be left floating.
- Modeling of the OUT pins takes the dropout voltage into account as per the section on Modeling LDO Linear Voltage Regulators.

# Modeling Negative Voltage Regulators

Negative voltage regulators generate a negative output voltage from a negative input voltage, but can often be enabled by either negative logic of positive logic, to make it easier to interface to its SHDN or EN pin to positive logic.

**Figure 3-48: Negative Voltage Regulator Simplified Diagram**




Sample data for modeling purposes:

- The SENSE pin (can also be ADJ or FB) has Vmax = -1.2V and Vmin = -1.25V.
- Vin range is -1.8V max and -30V min.
- Vout range is -1.22V max and min voltage is limited to the minimum input voltage + the dropout voltage.

The SHDN pin has the following positive/negative logic thresholds:

**Figure 3-49: Negative Voltage Regulator SHDN Specification**

| PARAMETER | CONDITIONS | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|
| ADJ Pin Bias Current | LT3015: $V_{IN}$ = –2.3V | –200 | 30 | 200 | nA |
| Shutdown Threshold | $V_{OUT}$ = Off-to-On (Positive) | 1.07 | 1.21 | 1.35 | V |
| | $V_{OUT}$ = Off-to-On (Negative) | –1.34 | –1.2 | –1.06 | V |
| | $V_{OUT}$ = On-to-Off (Positive) | 0.5 | 0.73 | | V |
| | $V_{OUT}$ = On-to-Off (Negative) | | –0.73 | –0.5 | V |

Using the sample data and the SHDN highlighted threshold specifications, the negative voltage regulator would be modeled as:

**Figure 3-50: Sample Model of Negative Voltage Regulator**

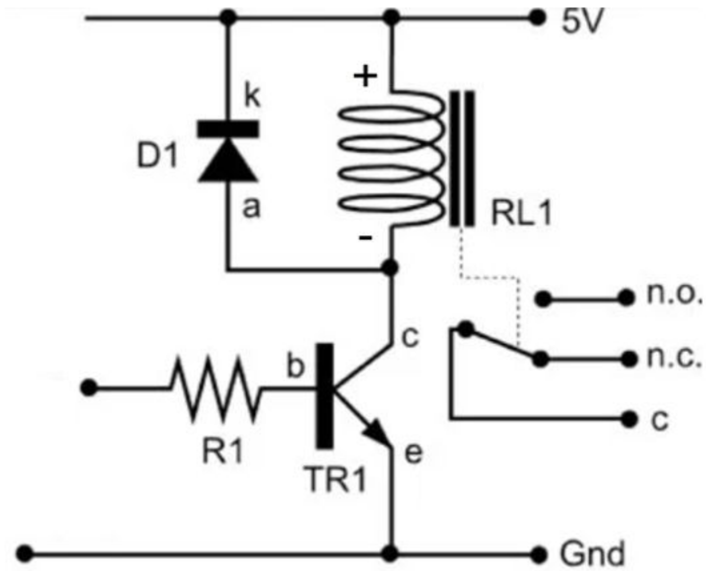
| # | | | PickList | Pin Treatment | | | Active Pin Characte | | | | For Power Pins | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| #Pin Number | Pin Name | Pin Desc | Pin Type | UP | SER | DN | Vin max | Vih | Vil | Vin min | Vmax | Vmin |
| 6 | ADJ | | FB | | | | | | | | -1.2 | -1.25 |
| 4 | GND | | GNDOUT | | | | | | | | 0 | |
| 5 | GND | | GNDOUT | | | | | | | | 0 | |
| 3 | SHDNn | Negative Logic | IN | | | A | GND | GND -0.5 | GND -1.34 | IN | | |
| #3 | SHDNn | Positive Logic | IN | A | | | 22 | 1.35 | 0.5 | 0 | | |
| 1 | IN | | PWR | | | | | | | | -1.8 | -30 |
| 2 | IN | | PWR | | | | | | | | -1.8 | -30 |
| 9 | IN | | PWR | | | | | | | | -1.8 | -30 |
| 7 | OUT | | PWROUT | | | | | | | | -1.22 | IN +0.5 |
| 8 | OUT | | PWROUT | | | | | | | | -1.22 | IN +0.5 |

- Negative logic shutdown is enabled, and positive logic shutdown is commented out, on SHDNn pin.
    - Negative logic shutdown has A in the DN column as the regulator is enabled when the input voltage is less than the Vil level.
    - Positive logic shutdown has A in the UP columns as the regulator is enabled when the input voltage is higher than the Vih level.
- Modeling of the OUT pins takes the dropout voltage into account as per the section on modeling LDO Linear Voltage Regulators.

# Modeling Coil Relays

A typical polarized coil relay circuit is shown in the figure below.

**Figure 3-51: Typical Polarized Coil Relay Circuit**




Polarized coil relays typically have the following coil operating characteristics, as shown in the figure below.

- Rated Coil voltage, and Max voltage
- Operate voltage
- Release Voltage
- Maximum Voltage
- Polarity

**Figure 3-52: Sample Relay Coil's Operating Characteristics**

| Rated Coil Voltage | Rated Current (mA) | Coil Resistance (W) | Must Operate Voltage (V) | Must Release Voltage (V) | Max Voltage (V) | Power Consumption (mW) |
|---|---|---|---|---|---|---|
| | | | % of Rated Voltage | | | |
| 3 VDC | 33.0 | 91 | 80% max. | 10% min. | 150% | Approx. 100 |
| 4.5 VDC | 23.2 | 194 | | | | |
| 5 VDC | 21.1 | 237 | | | | |
| 12 VDC | 9.1 | 1,315 | | | | |
| 24 VDC | 4.6 | 5,220 | | | | |

These characteristics essentially translate into Vin Max, Vih, and Vil levels for modeling purposes.

Due to the higher voltage and current requirements, driving the coil from digital logic is not feasible the relay coil is typically operated using a transistor that can tolerate the higher voltages and source/sink the higher currents as shown in Figure 3-51.

Since the relay coil could be floating (neither side connected to GND) and has thresholds, the best option to model the coil is as a partial FET, using only the Gate and Source pin types.

Sample data, for modeling purposes, of the relay version with 5 VDC operating voltage:

- Vin max = 5 * 150% = 7.5V
- Vih = 5 * 80% = 4V
- Vil = 5 * 10% = 0.5V
- Vin min = 0V
- Contact rating = 50V DC

If we put the above coil relay characteristics into a model, it would look as shown in the figure below.

**Figure 3-53: Sample Coil Relay Model**

| #Pin Number | Pin Name | Pin Desc | Pin Type | Pin Mate | Vin max | Vih | Vil | Vin min | Vo max | Voh | Vol | Vo min |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | COIL_P | | FETG | 2 | 7.5 | 4 | 0.5 | 0 | | | | |
| 2 | COIL_N | | FETS | | | | | | | | | |
| 3 | C | Common | GATE | 4 | 50 | | | 0 | 50 | | | 0 |
| 3 | C | Common | GATE | 5 | 50 | | | 0 | 50 | | | 0 |
| 4 | NO | Normally Open | GATE | 3 | 50 | | | 0 | 50 | | | 0 |
| 5 | NC | Normally Closed | GATE | 3 | 50 | | | 0 | 50 | | | 0 |

Modeling the coil relay in this fashion allows the analysis to check the coil thresholds and levels between the "+" and "-" coil pins, and the polarity of the coil.

# Modeling Quick Tips

### Modeling Power and GND pins

Sometimes data sheets, with many power and GND pins, number them sequentially as:

- VDD1, VDD2, VDD3, etc.
- VCCA1, VCCA2, etc.
- GND1, GND2, GND3, etc.

When it is known that they are internally physically the same power rail, model them as:

- VDDA
- VCCA
- GND

This will allow the analysis to check that component power and GND pins, with the same name, tied to the same internal power plane, are physically connected to the same voltage rail on the board.

# 4. Working With HyperLynx Schematic Analysis Libraries

The HyperLynx Schematic Analysis tool requires a library database of component models to perform an analysis.

As described in **Model Template Contents**, each model contains a manufacturer part number, description, pin numbers, pin names, and the electrical characteristics of each pin. You can add, edit, and remove component models from the database using the model library. One or more libraries can be enabled at a time for a project.

Some organizations prefer to have a specific individual manage the model library while restricting access to other users for non-disclosure agreements (NDA) and controlled goods purposes. License restriction enables this control.



## Configuring a Model Library Database

You specify the location from which the HyperLynx Schematic Analysis tool accesses intelligent models.

The HyperLynx Schematic Analysis tool stores models in three different database locations: as part of the project file, as a local file or a file located on a server, and as part of the HyperLynx Schematic Analysis online model database. The tool searches these databases simultaneously when searching for components in the model library.

You can identify the source of a model by looking at the model's Origin column value in the Model Library dialog box (from the Global bar, click **Model Library** > **Open Model Library**): PROJECT, FILE or SERVER, or ONLINE.

**Table 4-1: Database Sources**

| Origin | Description |
|---|---|
| PROJECT | The tool adds a model to the project database when you assign a model to a component in the Assign Part dialog box (click **BOM** tab, right-click a component and choose **Assign**). At that time, the model is copied from one of the databases (FILE, SERVER, or ONLINE) into the project file database. This enables you to share and send models stored in the project file. |

**Table 4-1: Database Sources (continued)**

| Origin | Description |
|---|---|
| | Working With BOMS in the *HyperLynx Schematic Analysis Guide and Reference*. |
| FILE or SERVER | You specify a SQL database as either a SQLite database file, or a MySQL or PostgreSQL database file running in a server environment. See Creating a Model Library Database in the *HyperLynx Schematic Analysis Guide and Reference*. |
| ONLINE | The HyperLynx Schematic Analysis online model library database is delivered as part of the HyperLynx Schematic Analysis tool. |

## Procedure

1. In the HyperLynx Schematic Analysis tool, on the Global bar, click **Settings** > **Library** category. The Library group displays.

   In the **Library** category, click **Configure**, the Library Settings dialog box opens.

2. Configure the intelligent model library database in any of the following ways:

<table>
<tr><th>If you want to:</th><th>Do the following:</th></tr>
<tr><td>Use an existing local model library database</td><td>Note<br>The HyperLynx Schematic Analysis installation process installs an empty local model library database (Untitled_Components_Database.vcdb).<br><br>In the Local Model Library subgroup, do the following:<br>1. Verify the File option is selected in the Database Select field, then do either of the following:<br>• To use the default model database created during installation of the HyperLynx Schematic Analysis tool, confirm the pathname in the Filename text entry box.<br>• To use a different local file model database, click Browse, navigate to and select a different database file (.vcdb) in the Open file dialog box, and then click Open.<br>2. (Optional) Click the Test Connection button to verify that the tool can connect to the specified database.<br><br>You can create additional local model library databases as needed. For any local model library database, you must create and add the models that you</td></tr>
</table>

<table>
<tr><th>If you want to:</th><th>Do the following:</th></tr>
<tr><td></td><td>need. For more information on creating and adding models to a local model library database, refer to Manipulating the Contents of the Model Library.</td></tr>
<tr><td>Create and use a new local model library database</td><td>In the Local Model Library subgroup, do the following:<br>1. Verify the File option is selected in the Database Select field.<br>2. Click Create New and, in the Create Model Database dialog box, navigate to the directory where you want the new model database located, type a name for the database in the File name field, and then click Save.<br>The tool populates the Filename field with the path to the new database file (.vcdb).<br>3. (Optional) Click the Test Connection button to verify that the tool can open the specified local database.<br>You can create additional local model library databases as needed. For any local model library database, you must create and add the models that you need. For more information on creating and adding models to a local model library database, refer to Manipulating the Contents of the Model Library.</td></tr>
<tr><td>Use an Enterprise-level server model library database</td><td>In the Local Model Library subgroup, do the following:<br>1. In the Database Select field, select either the Mysql or Postgresql option from the dropdown list.<br>2. Enter your values for the Address, Database Name, Username, Password, and Port fields.<br>3. (Optional) Click the Test Connection button to verify that the tool can open the specified local database.</td></tr>
<tr><td>Use the HyperLynx Schematic Analysis online model library database</td><td>Close the Library Settings dialog box to return to the Settings view. In the Online Model Library subgroup, do the following:<br>1. Verify that the Enable Online Access switch is enabled.<br>2. Specify whether you want to use a proxy to communicate with the online server by doing one of the following:<br>• To use a Proxy — Select the Enable Proxy option to enable it and, then, specify the proxy’s Port address, Host (IP address), Username, and Password.<br>• To not use a proxy — Leave the Enable Proxy option disabled, and leave the Port, Host, Username, and Password fields empty.<br>Tip<br>If you have IT restrictions that block your ability to communicate with the online server, make sure that you are not blocking communications to valydate-api.com through port 443.</td></tr>
</table>

| If you want to: | Do the following: |
|---|---|
| | 3. (Optional) Click the **Test Connection** button to verify that the tool can communicate with the specified port. |

3. Exit the HyperLynx Schematic Analysis tool, and then restart it for your changes to take effect.

# Manipulating the Contents of the Model Library

After opening the model library, you can add, import, update, search for, delete, and edit models.

## Prerequisites

- You have a configured model library. See **Configuring a Model Library Database**.
- For each component in the bill of materials for the schematic there is a completed model saved in the project folder. See **Creating HyperLynx Schematic Analysis Models** for information on accessing the model templates, completing them, and saving them.
- For each type of bulk component (capacitors, resistors, and so on), there is a spreadsheet listing the parts, and it is saved in the project folder. See **The Bulk Component Model Template**.
- The HyperLynx Schematic Analysis tool is open.

## Procedure

1. On the Global bar, click **Model Library** > Model Library group > **Open Model Library** .
2. In the Model Library dialog box, do any of the following:

| If you want to... | Do the following: |
|---|---|
| Add a new component | 1. From the Global bar, click **Model Library** > Model Library group > **Import Model** to import a model that is an intelligent model file (*.xls*, *.xlsx*, or *.vcpt*).<br>The Select Model file dialog box opens.<br>2. In the dialog box, select the desired model file(s) and click **Open**.<br>The Model Library dialog box opens to display the models.<br>If a model with the same manufacturer part number and customer part number already exists in the library, an Error dialog box displays.<br>3. (Optional) If an Error dialog box displays, do one of the following: |

| If you want to... | Do the following: |
|---|---|
| | • To overwrite the preexisting model in the Library with the new model, click the **Update Model** button.<br>• To overwrite all selected preexisting models in the Library, click the **Update All** button.<br>• To cancel the model import with no changes, click the **Drop Model Import** button. |
| Search for a model | 1. In the Search text entry field of the Model Library dialog box (see Add a new component row in this table), type a manufacturer part number or customer part number.<br>**Note**<br>Part names cannot include special characters. If a part name does have special characters, you should convert the special characters to underscores.<br>2. Click the **Search** button.<br>A list displays of all the models in the library that match the characters in the search field.<br>**Note**<br>You can refine your search by clicking the **Search Options** ≡ button to access additional search options, and the **Filter** button to filter on model sources. |
| Delete a component | In the Model Library dialog box (see Add a new component row in this table), right-click on the model you want to delete, and choose **Delete**.<br>The tool removes the model from the library and removes the entry from the list of models. |
| Edit a component | 1. In the Model Library dialog box (see Add a new component row in this table), right-click on the model you want to edit, and choose **Edit**.<br>The Edit Model dialog box displays, enabling you to view and edit a model that exists in the model library database.<br>2. After editing the model, click **Update**.<br>The tool reports that the model you edited is now *app specific* in the project. |

| If you want to... | Do the following: |
|---|---|
| Duplicate a component | 1. In the Model Library dialog box (see Add a new component row in this table), right-click on the model you want to edit, and choose **Duplicate**.<br>A Duplicate Model dialog box displays.<br>2. Enter a name for the new model in the Model Name text entry field, and then click **OK**.<br>The tool adds the new model to the bottom of the list of models. |
| Export a component | 1. In the Model Library dialog box (see Add a new component row in this table), right-click on the model you want to export, and choose **Export**.<br>An Export Model dialog box displays.<br>2. Navigate to the location to which you want to export the model.<br>3. In the Export Model dialog box, enter a name for the exported model in the File name field, and specify the desired format for the exported model (*.*xlsx* or *.*vcpt*); then, click **Save**.<br>The tool exports the model to the specified location. |

# Third-Party Information

Details on open source and third-party software that may be included with this product are available in the *<your_software_installation_location>/legal* directory.